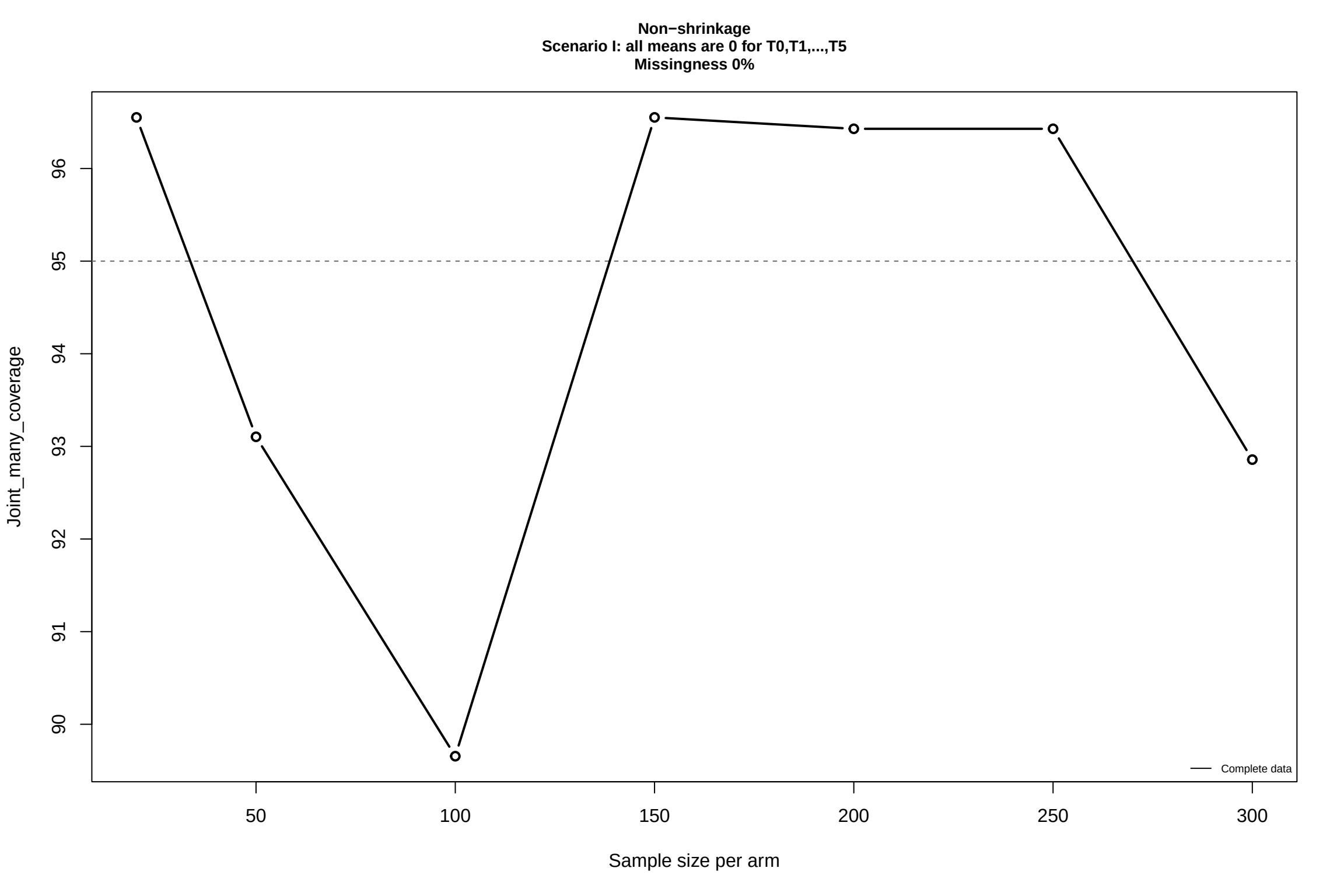
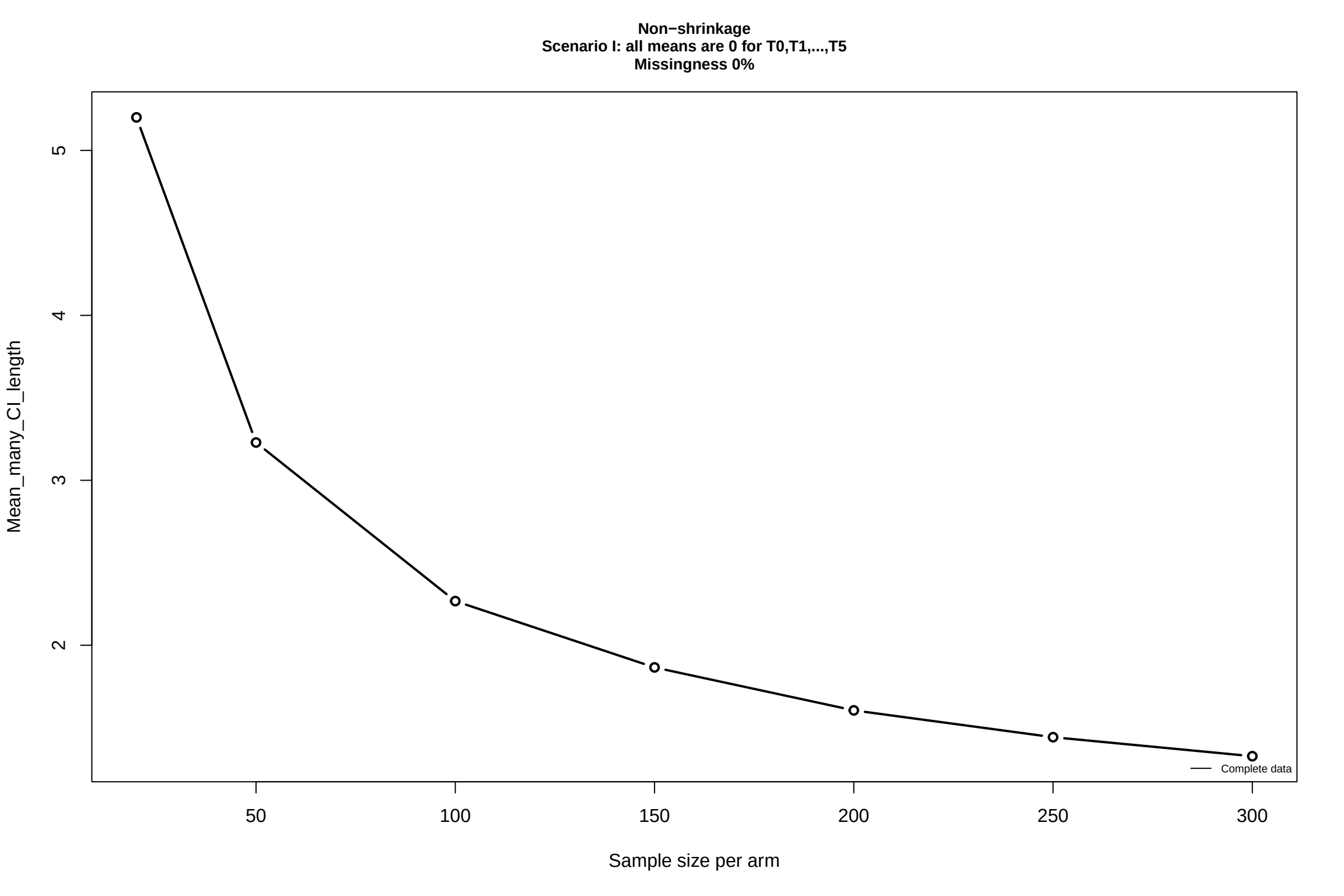
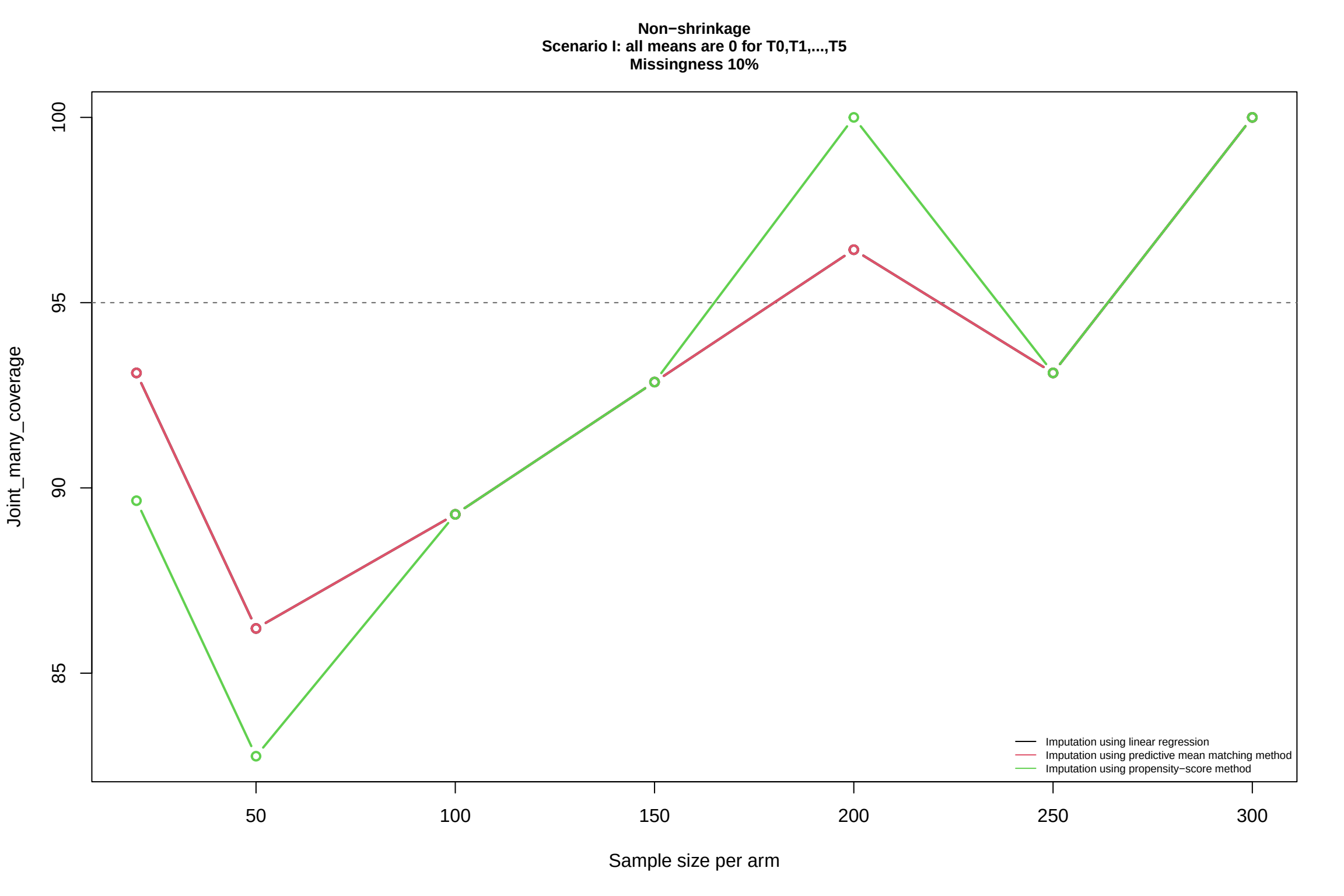
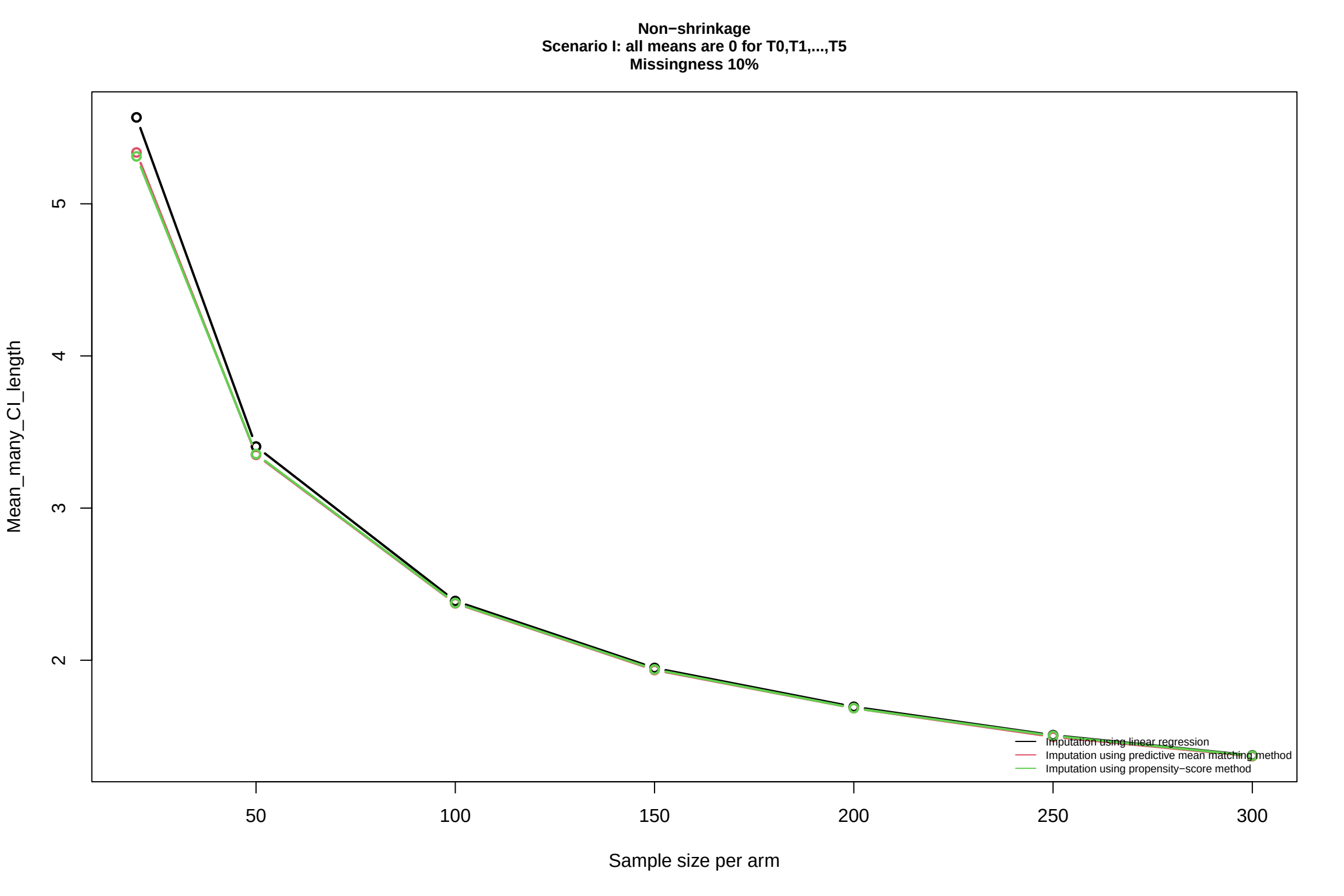
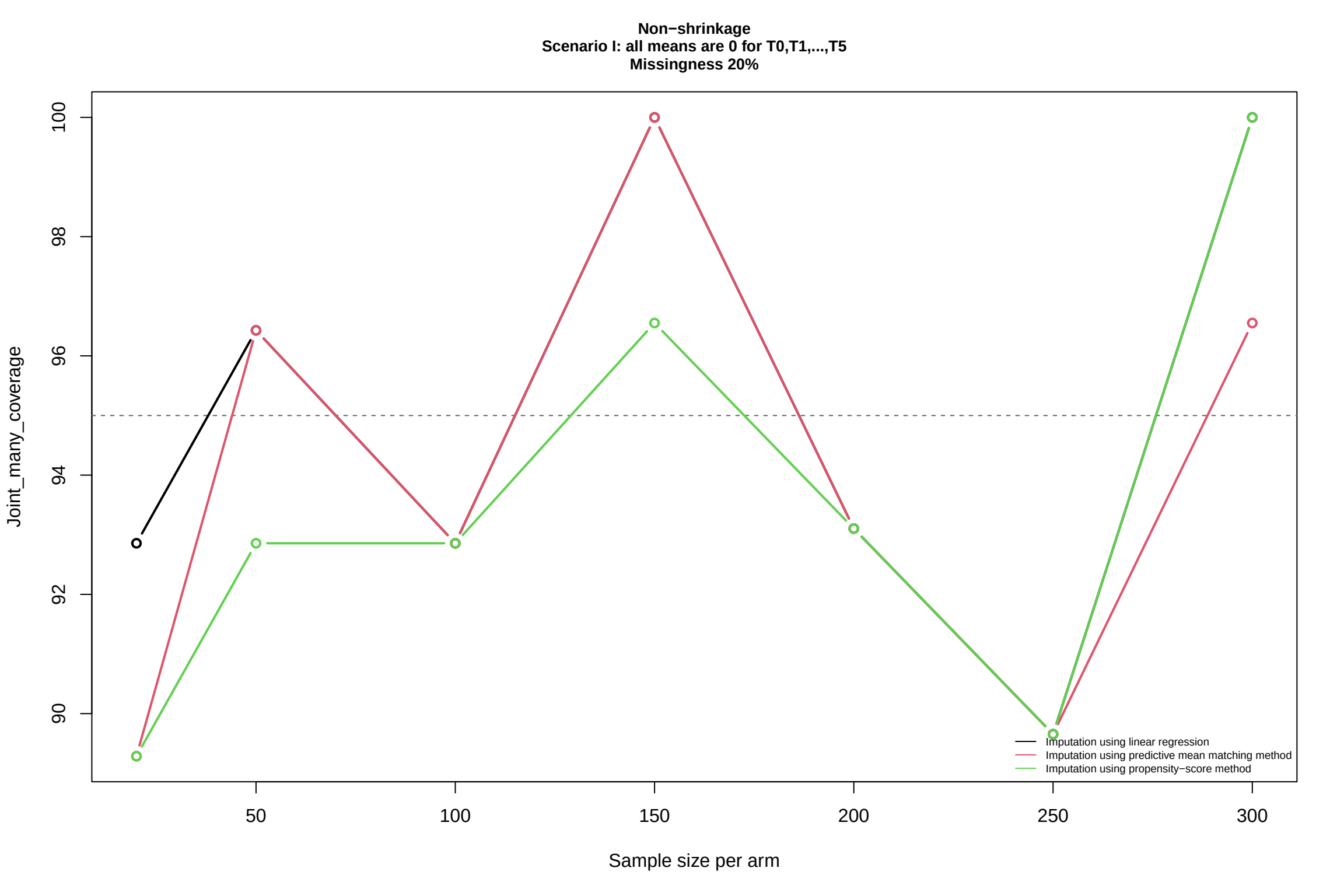
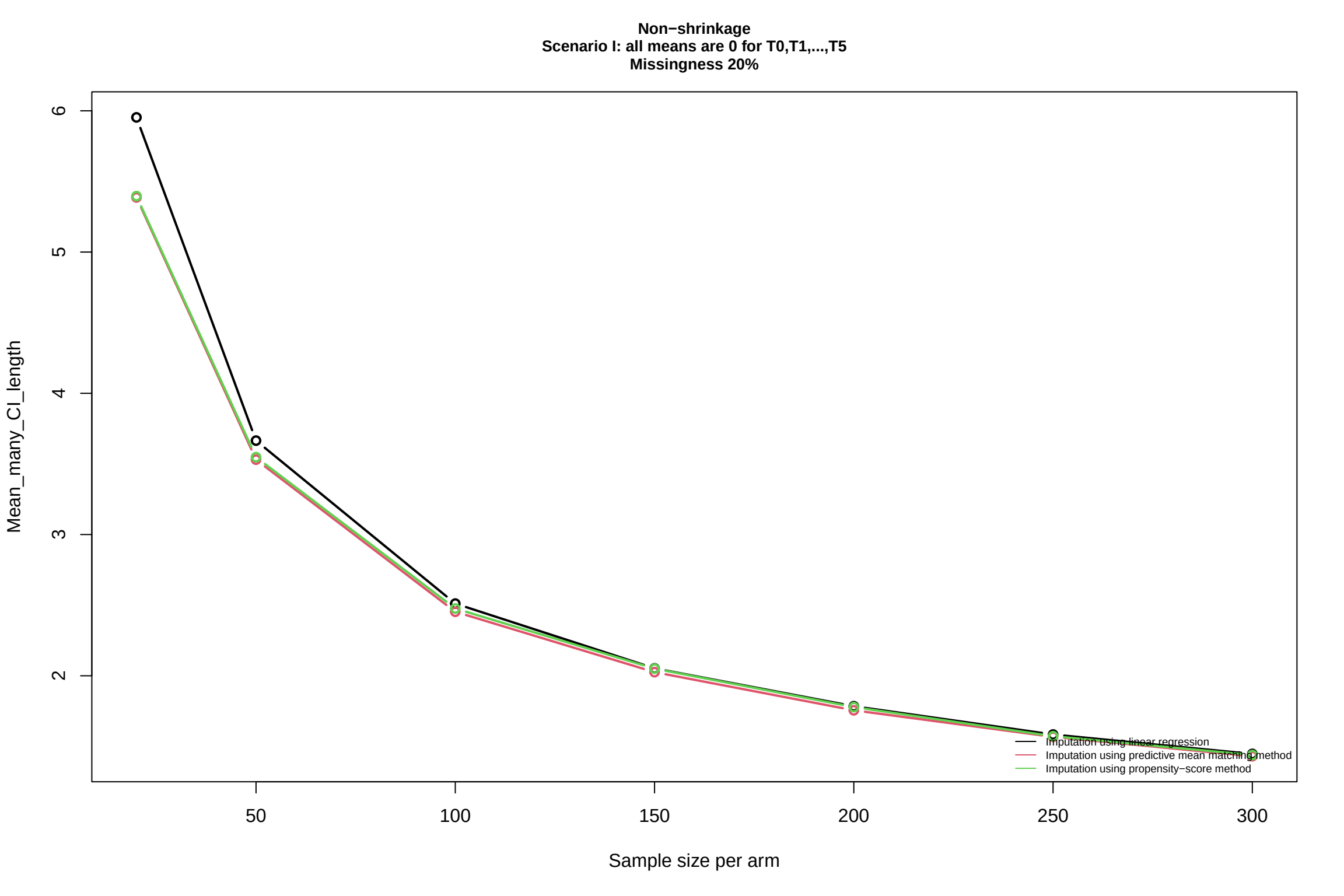

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 0%
Joint_many_coverage
90
91
92
93
94
95
96
50
100
150
200
250
300
Sample size per arm
Complete data

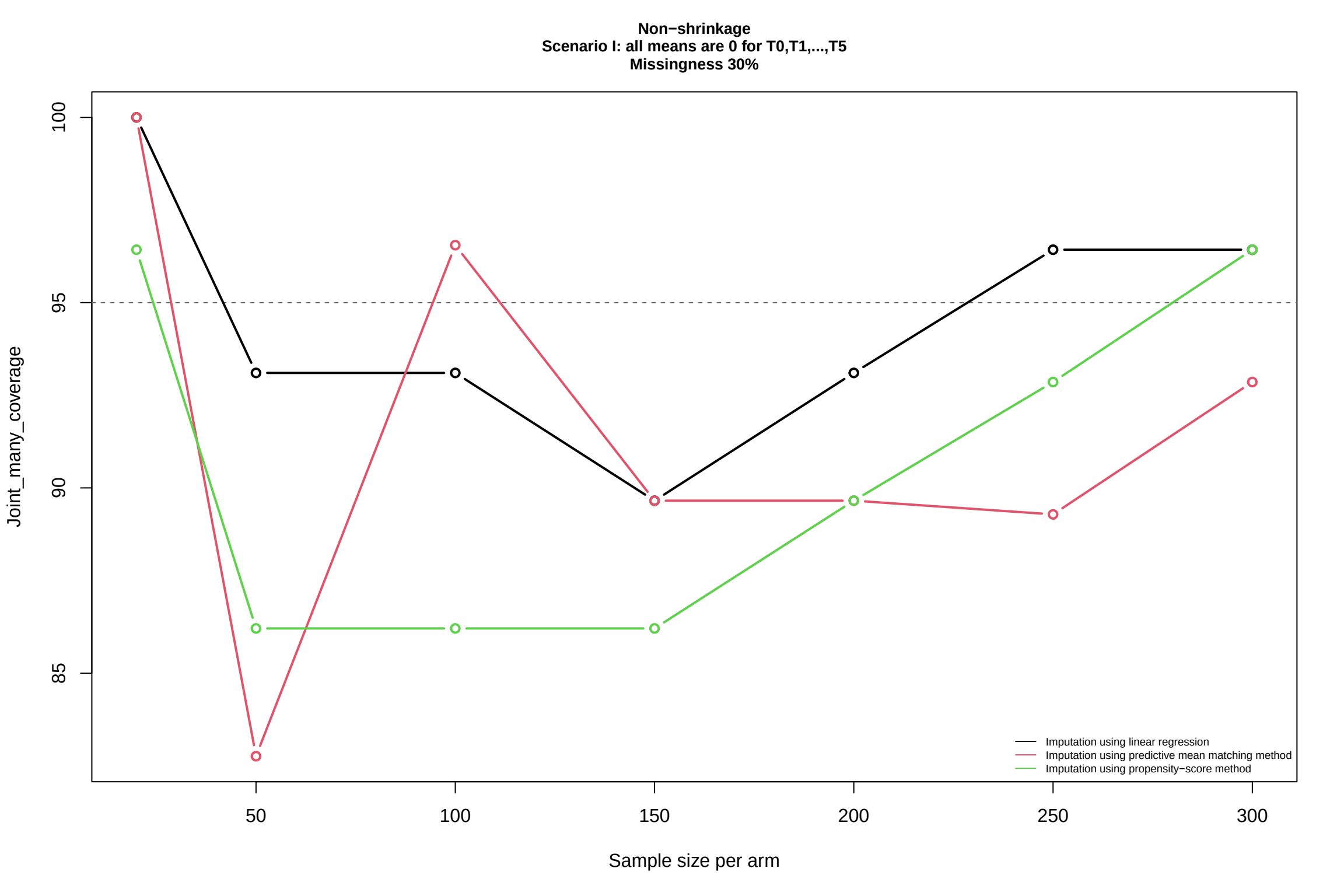

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

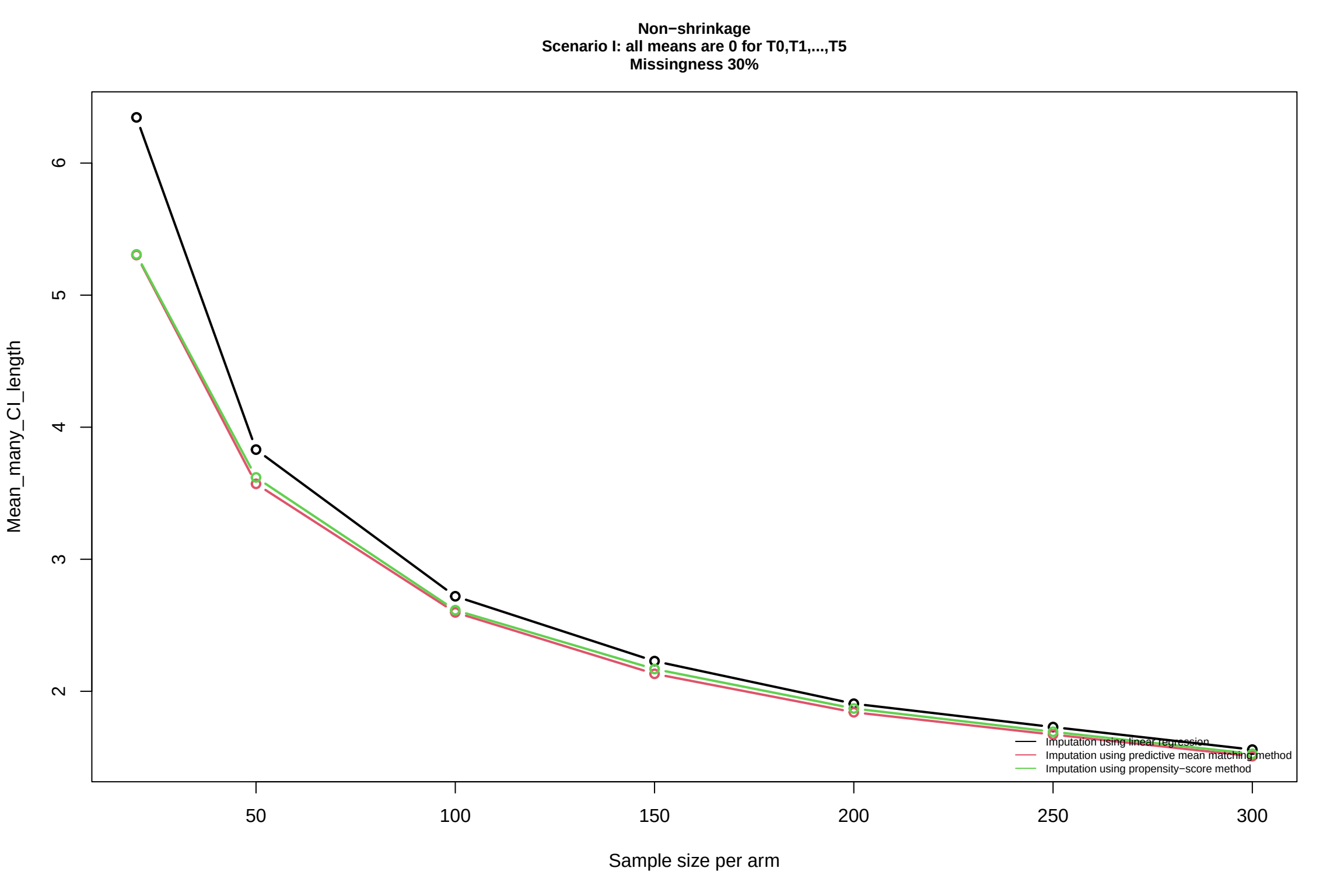

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 10%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

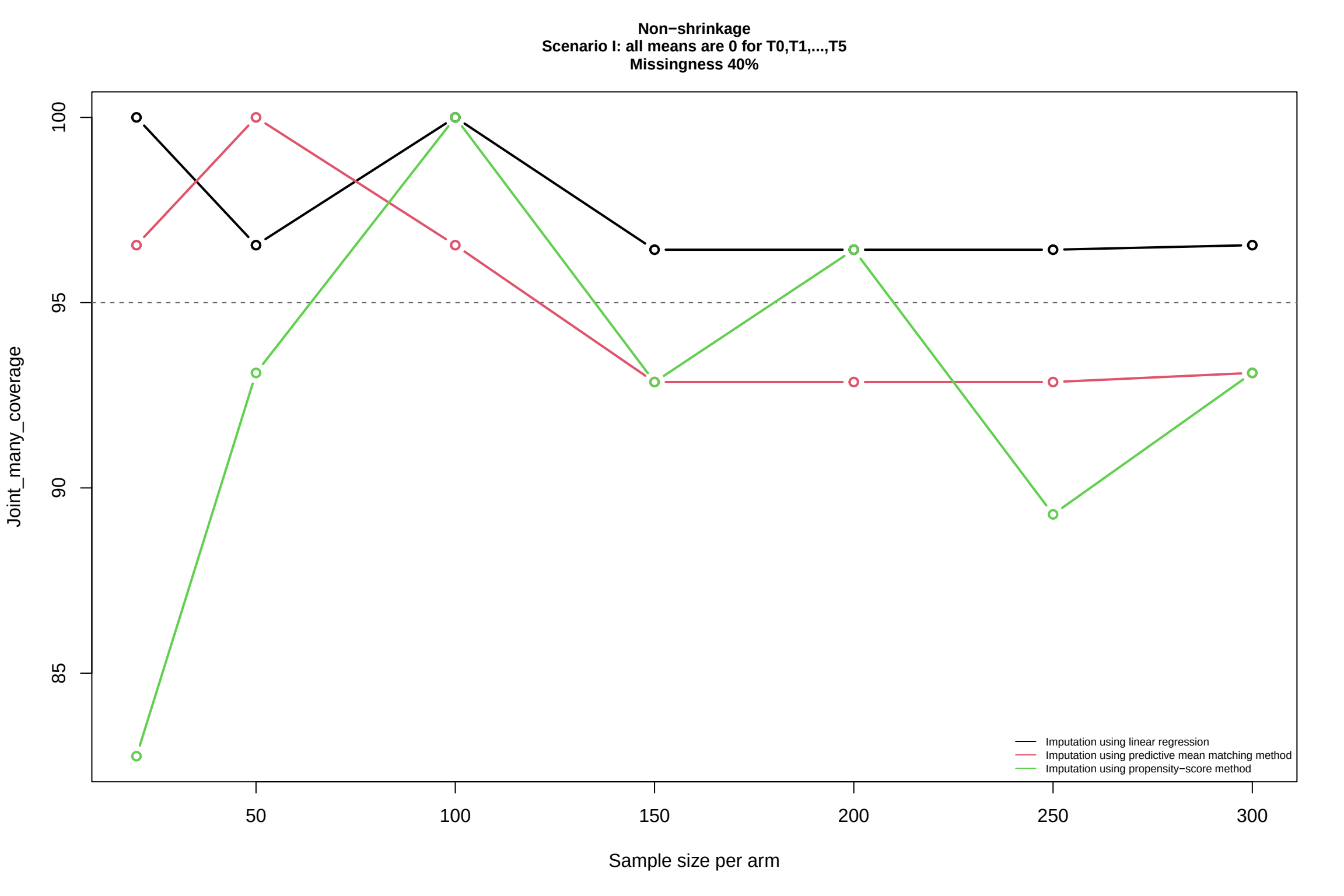

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 10%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

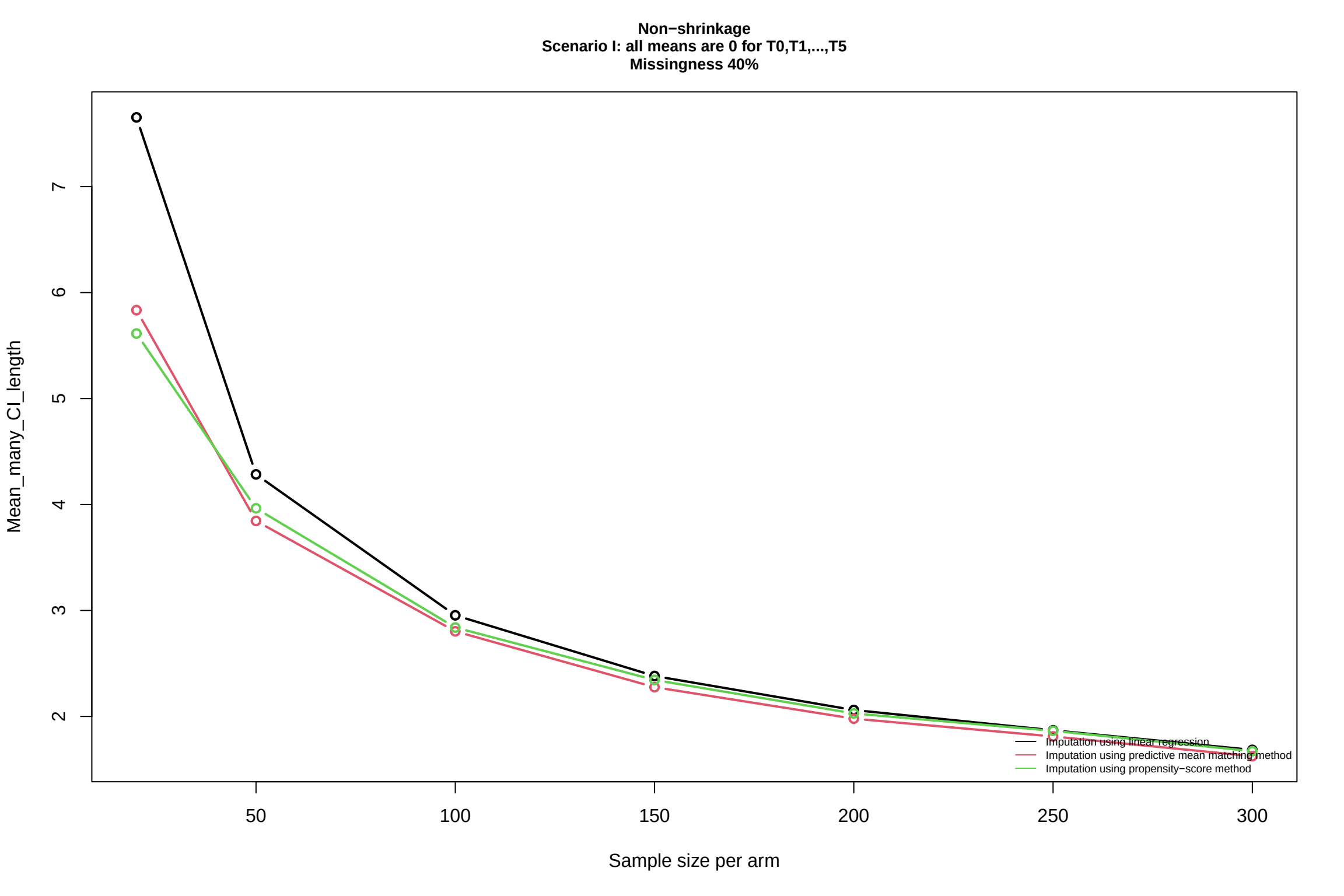

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 20%
Joint_many_coverage
100
98
96
94
92
90
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

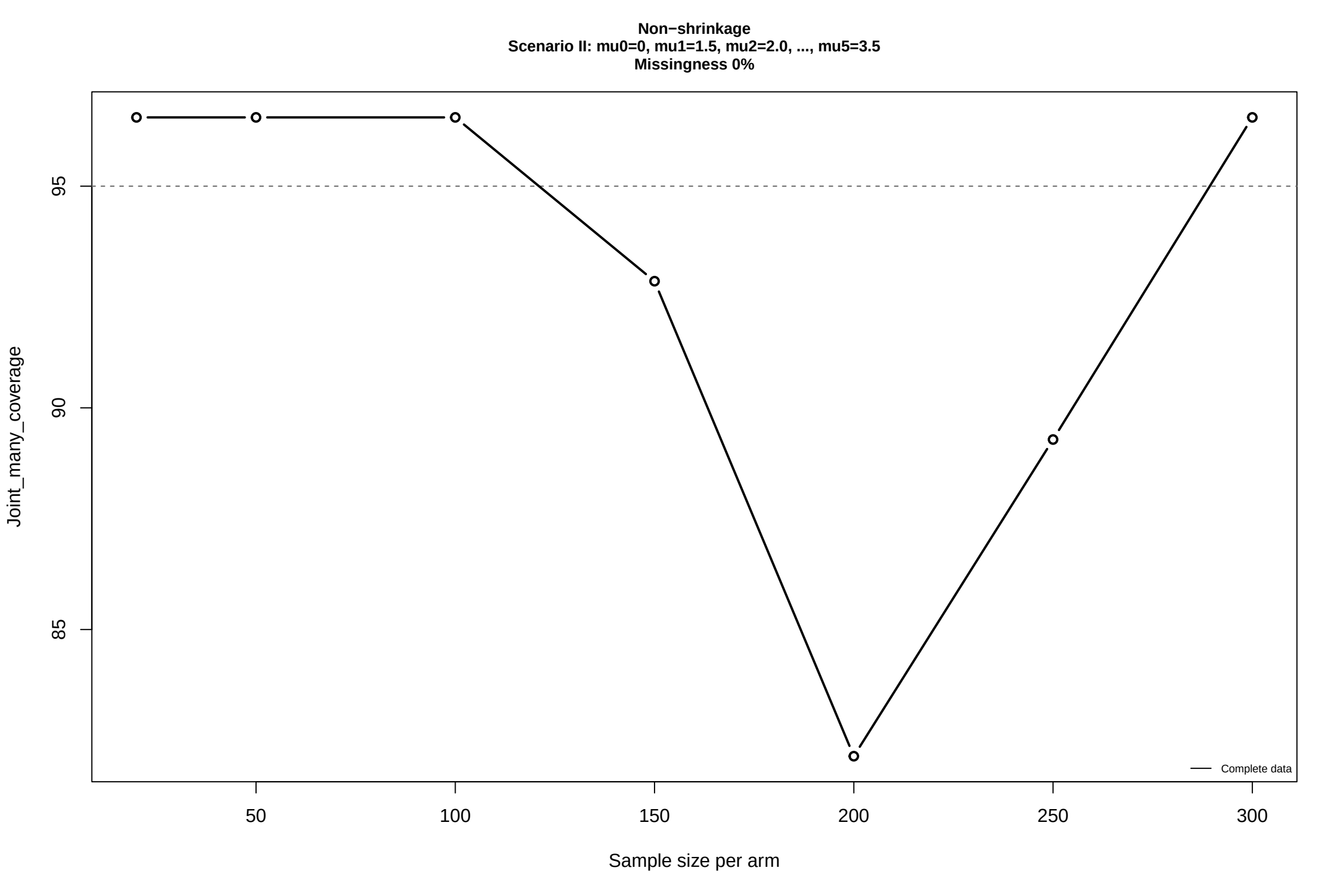

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 20%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

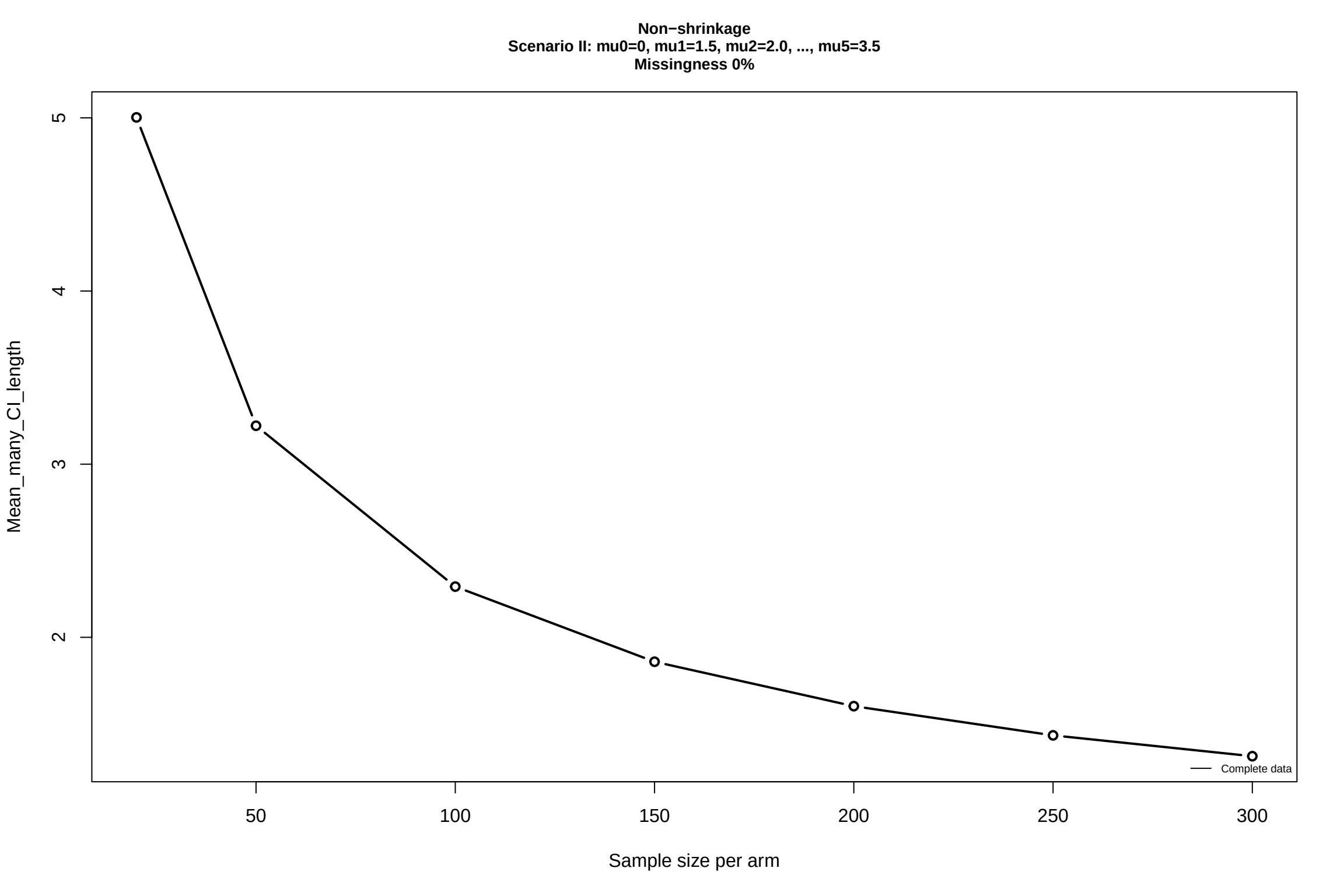

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 30%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

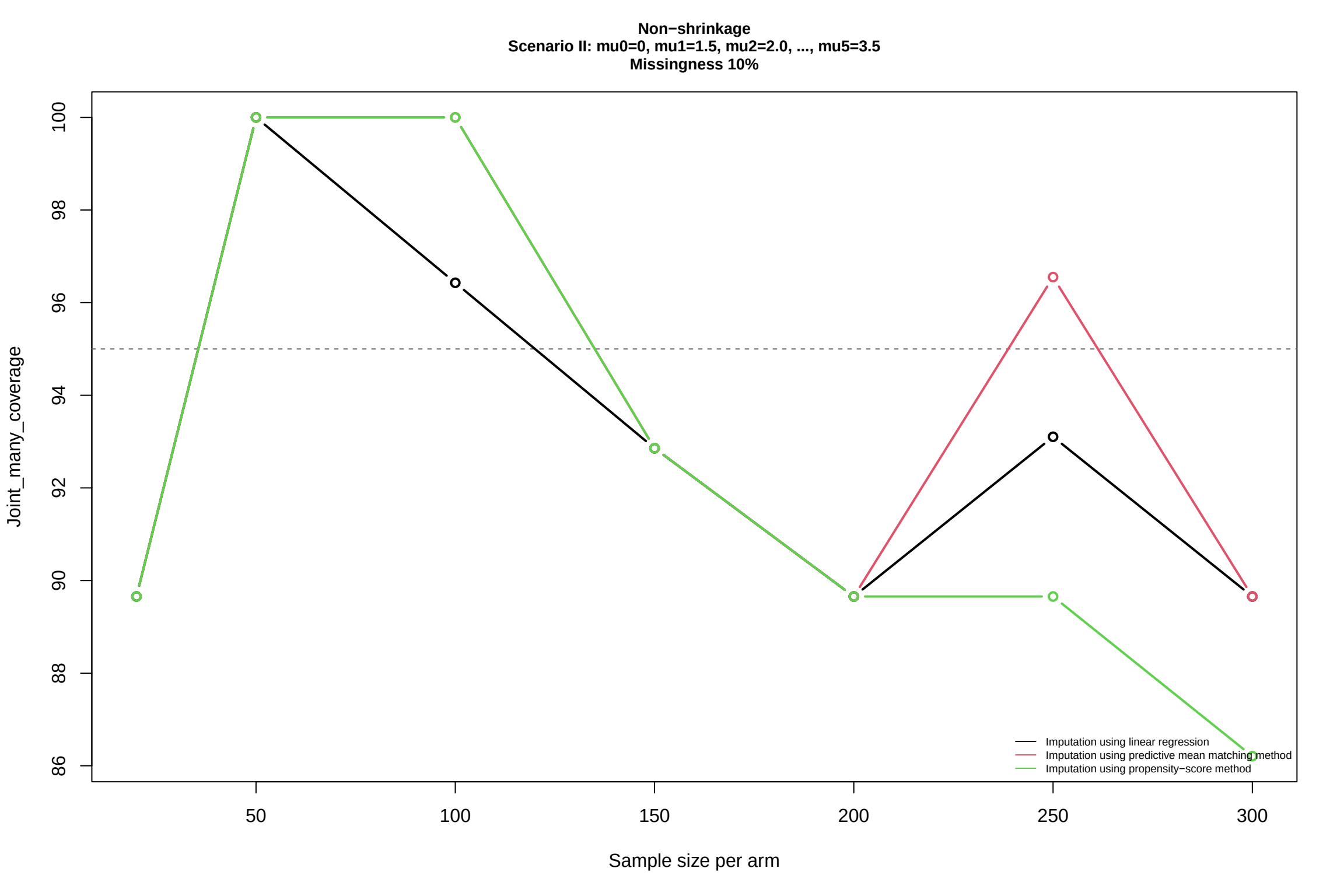

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 30%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

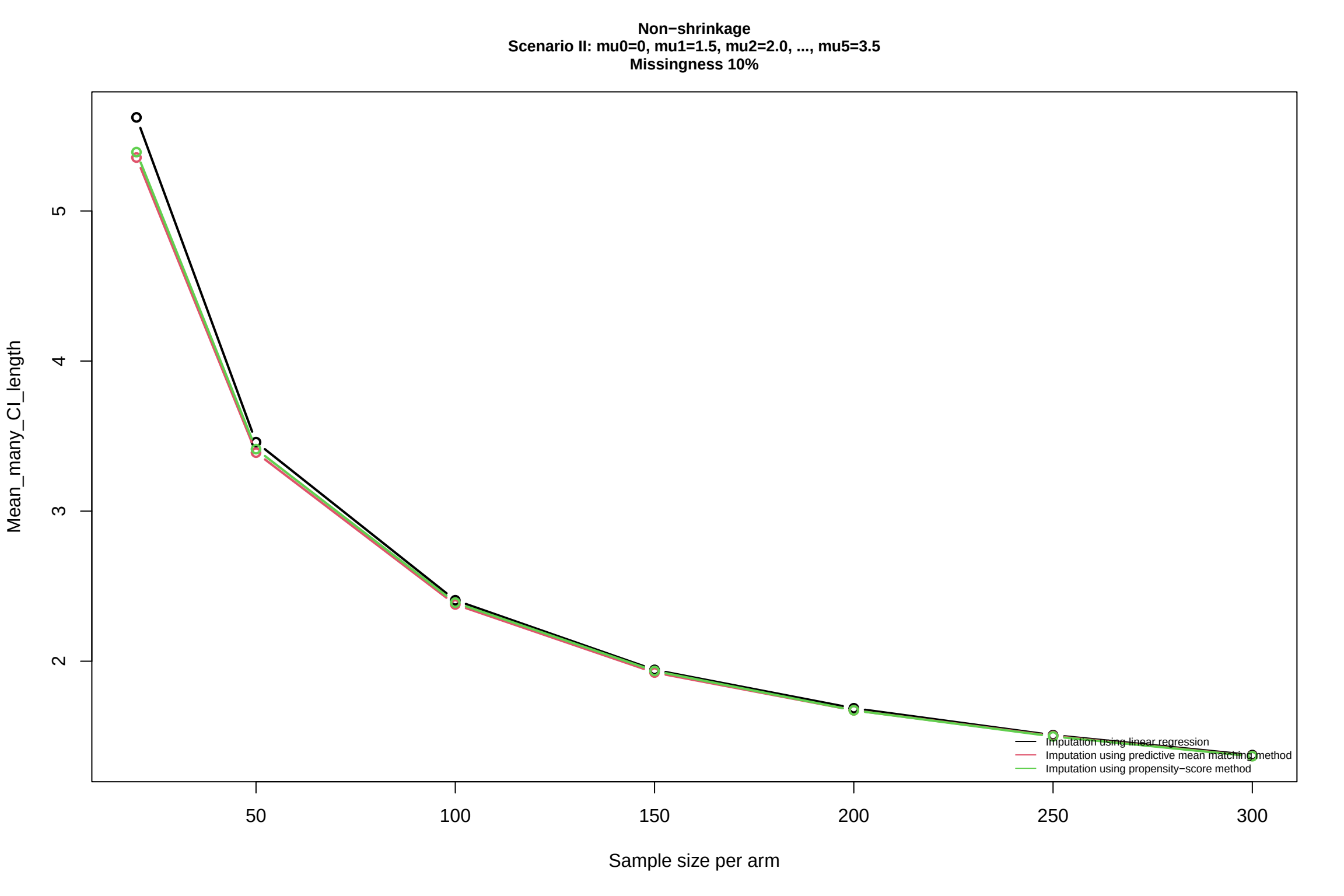

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 40%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

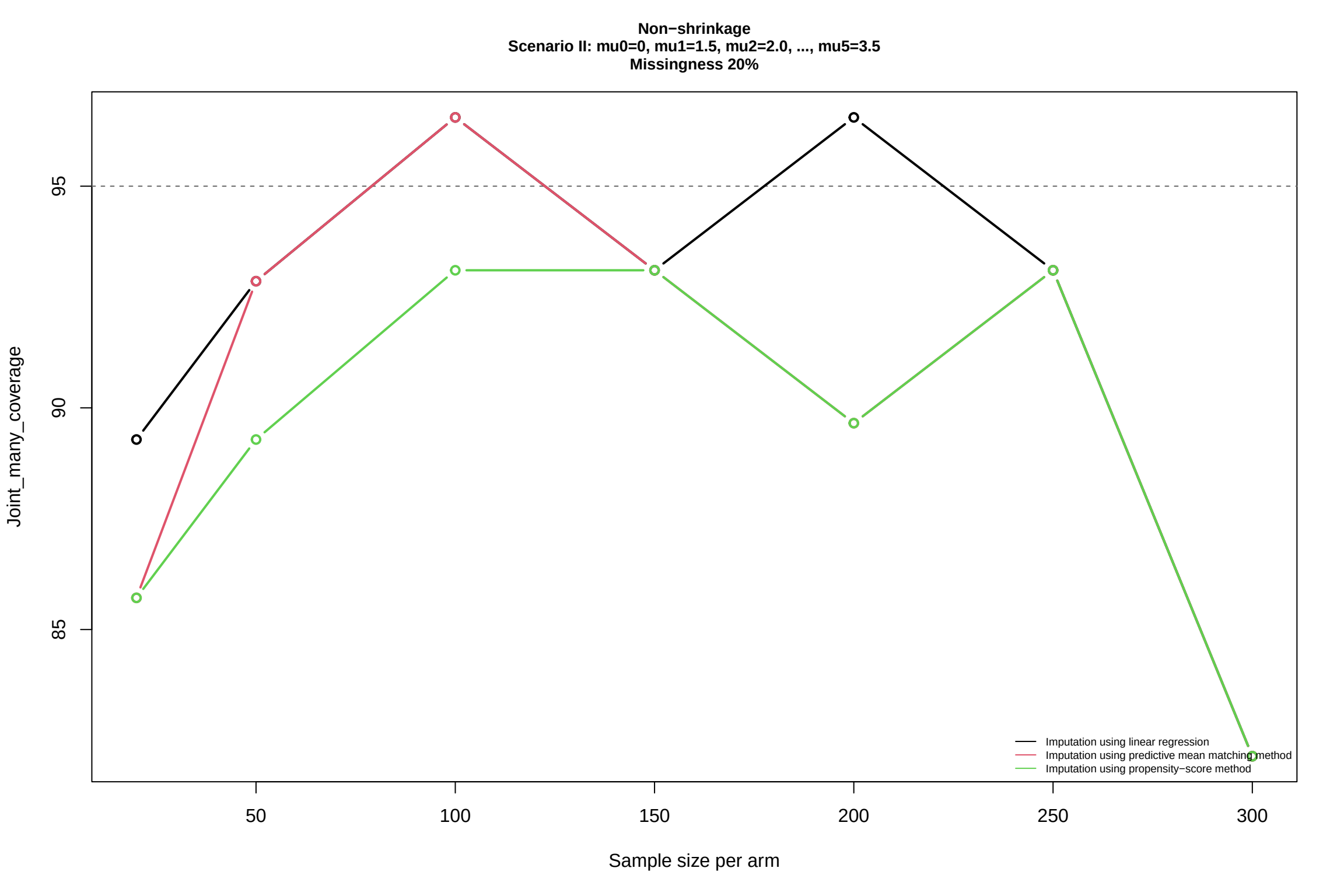

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 40%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

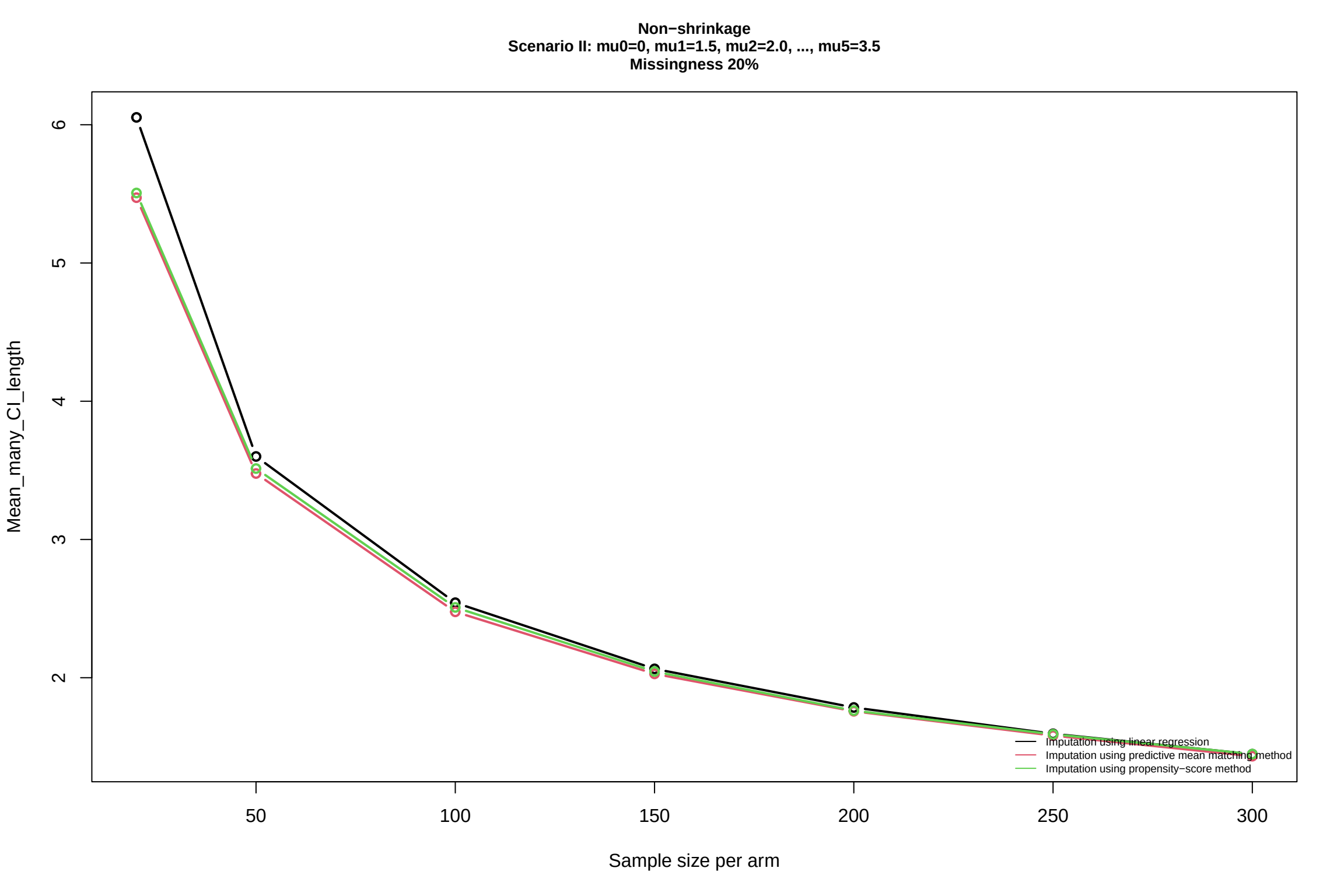

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 0%
Joint_many_coverage
95
90
85
50
100
150
200
250
300
Sample size per arm
Complete data

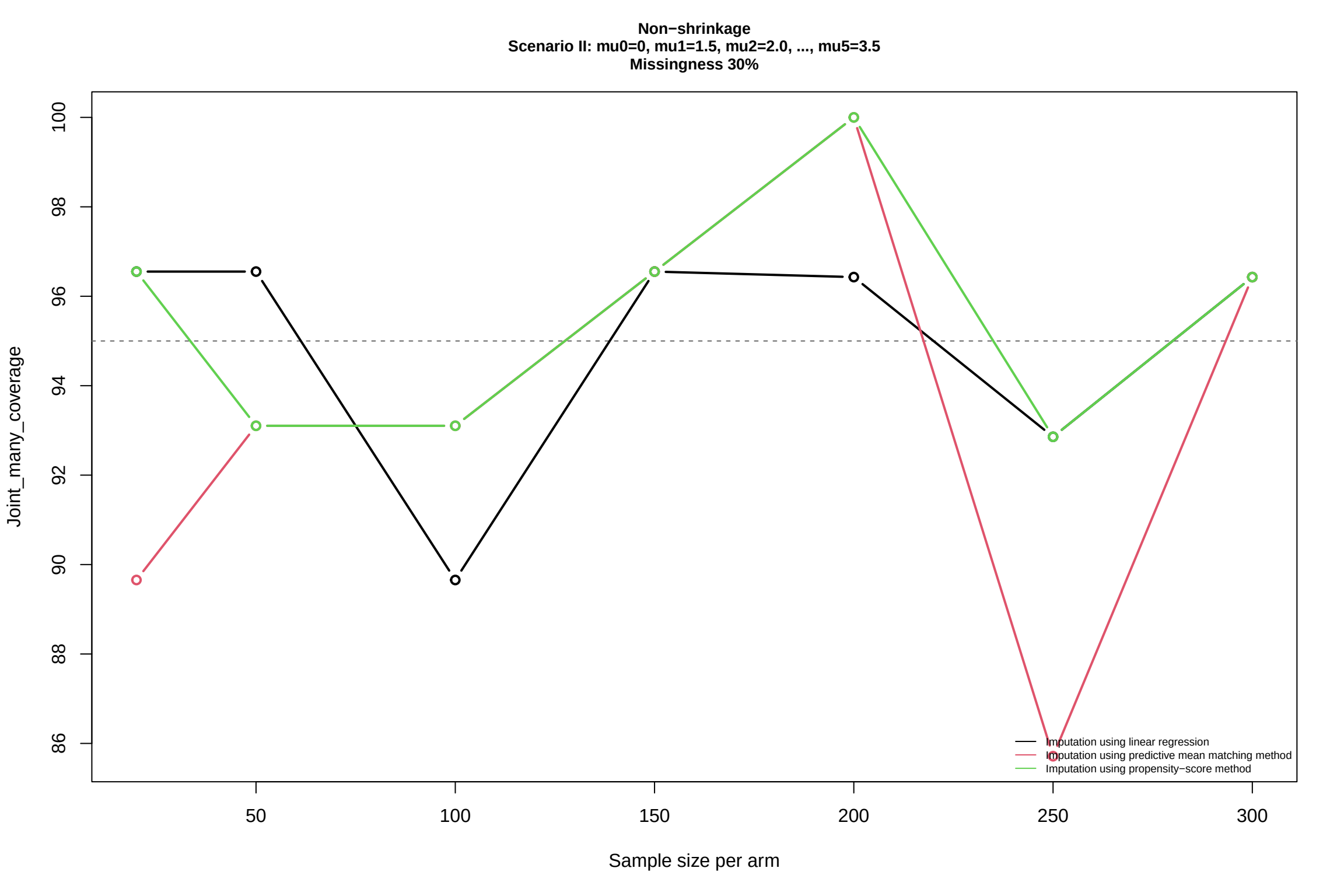

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

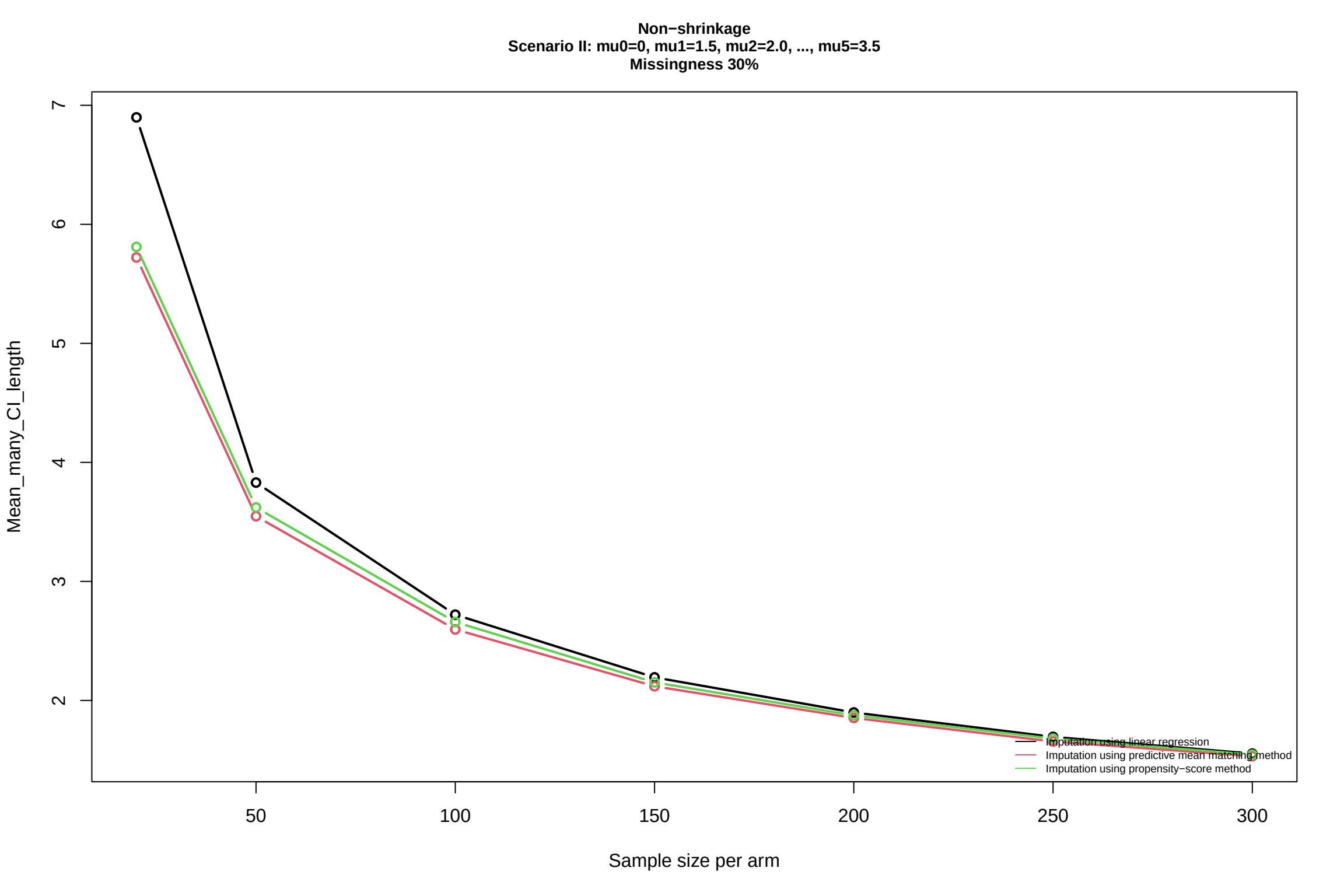

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 10%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

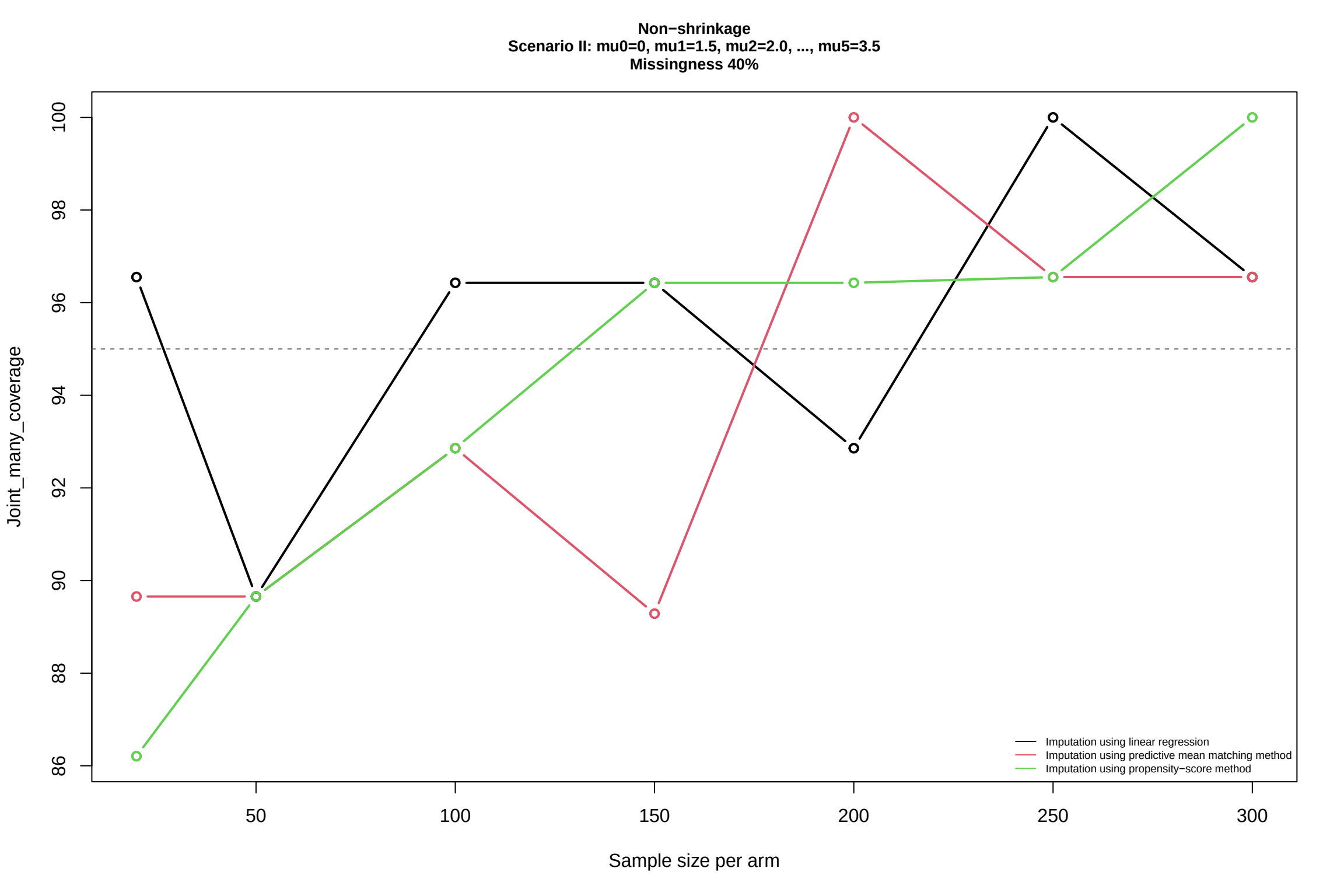

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 10%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

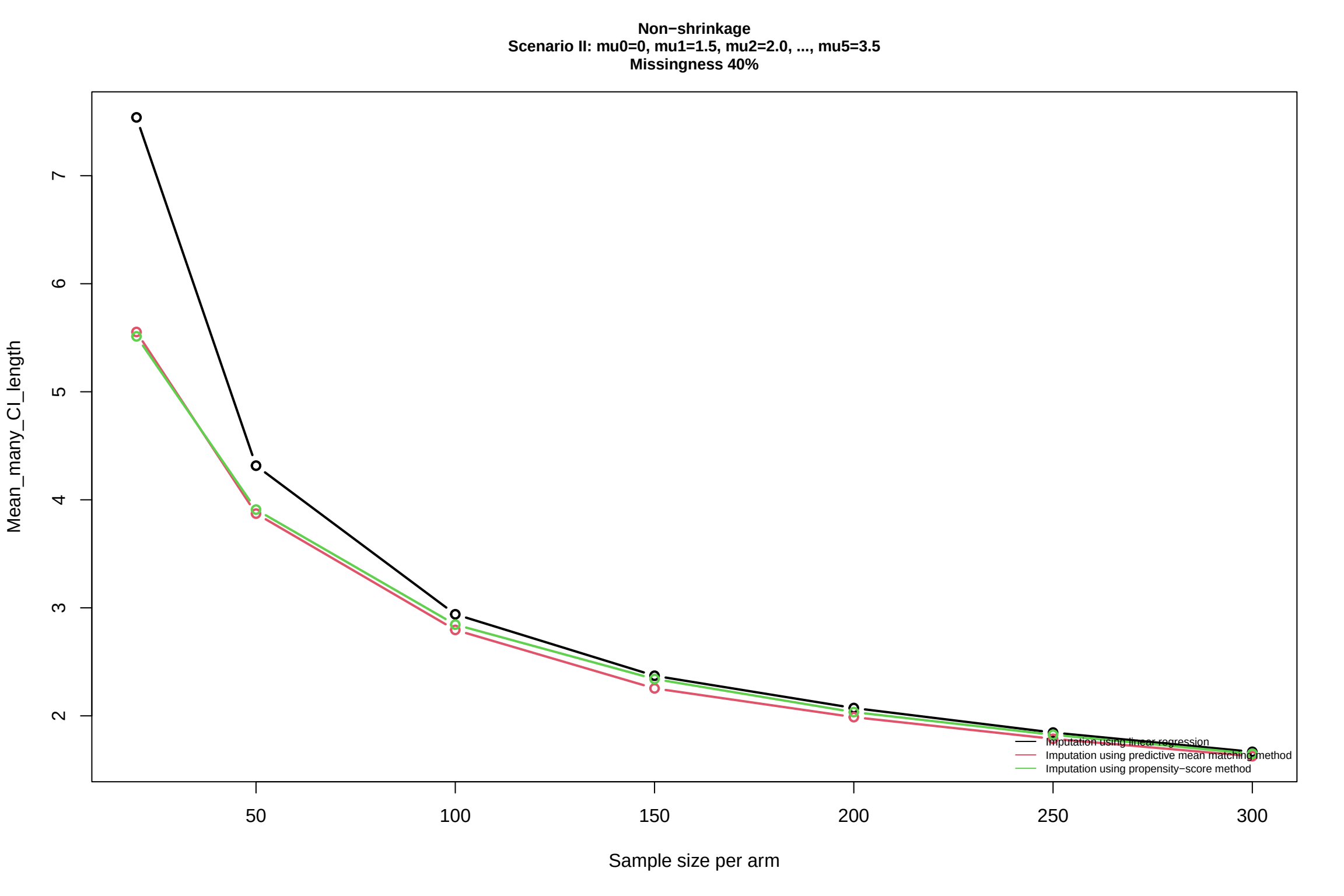

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 20%
Joint_many_coverage
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

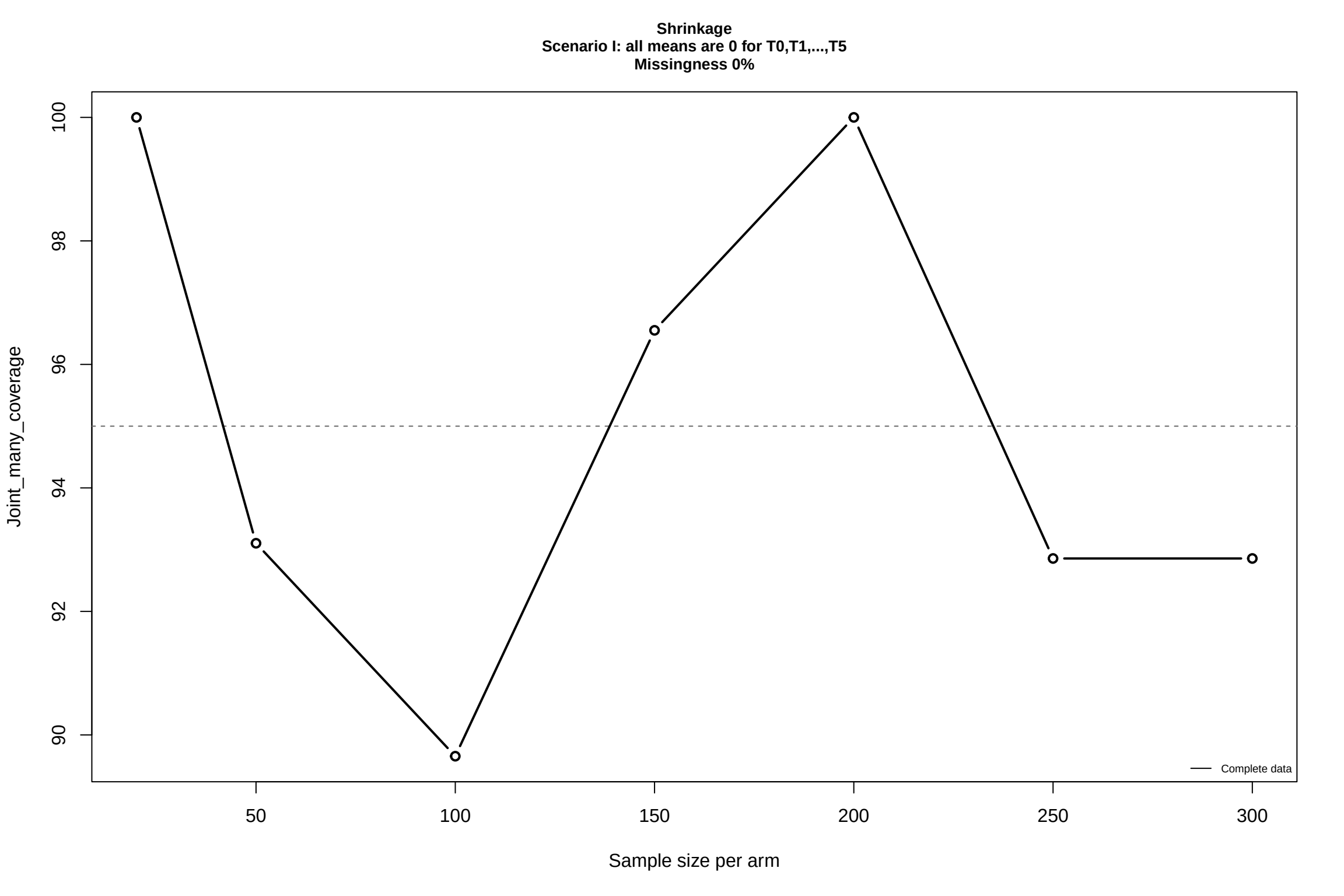

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 20%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

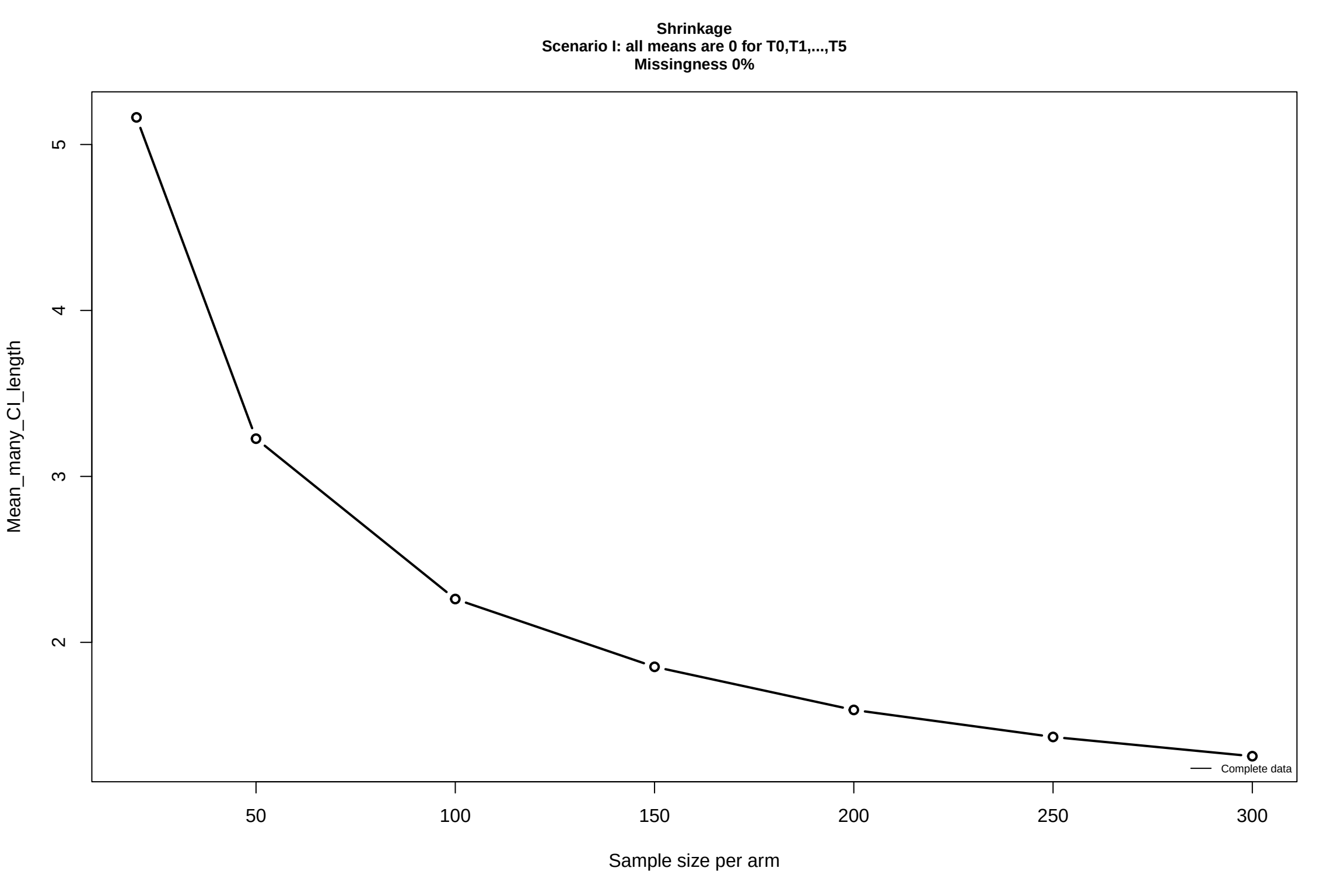

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 30%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

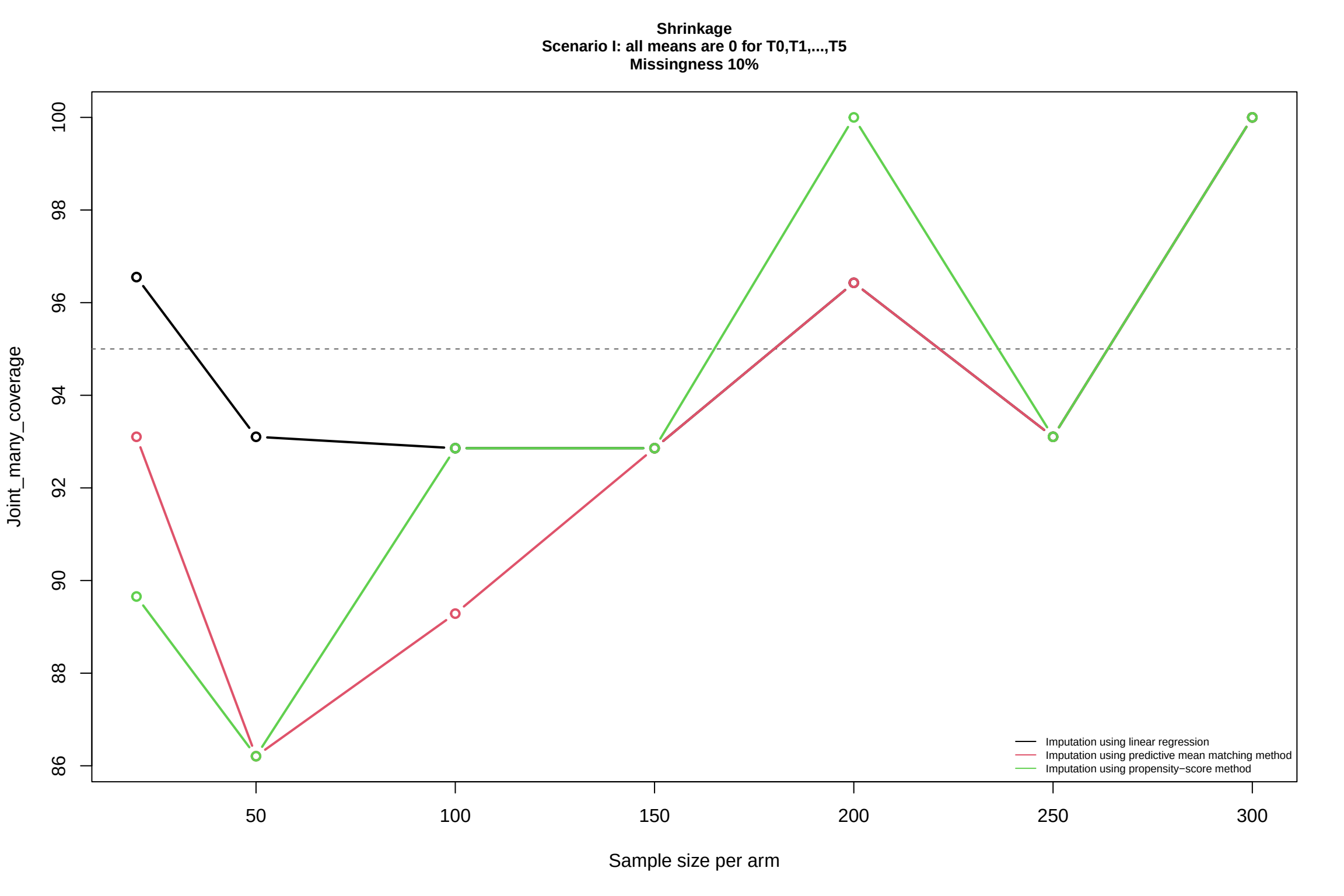

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 30%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

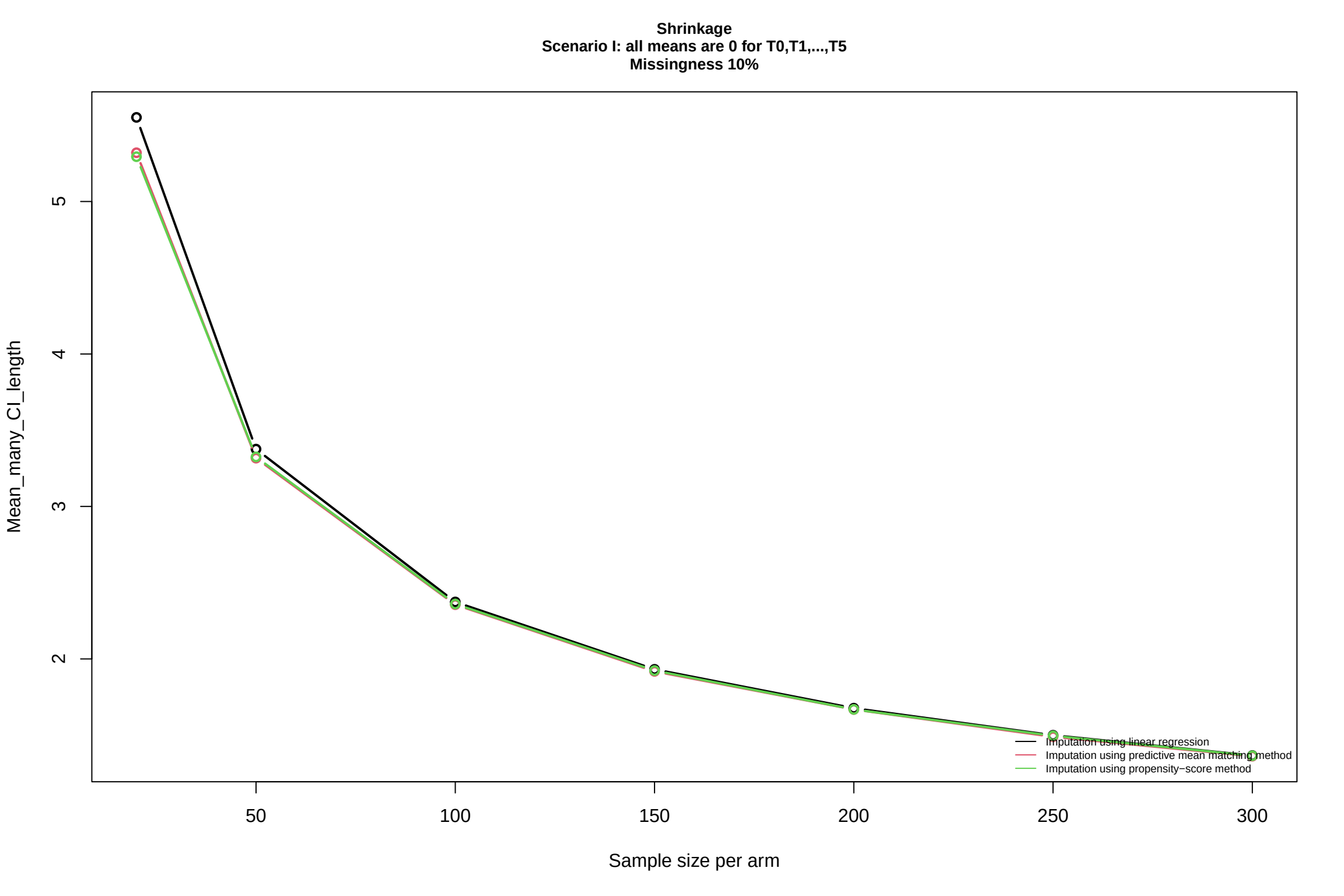

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 40%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

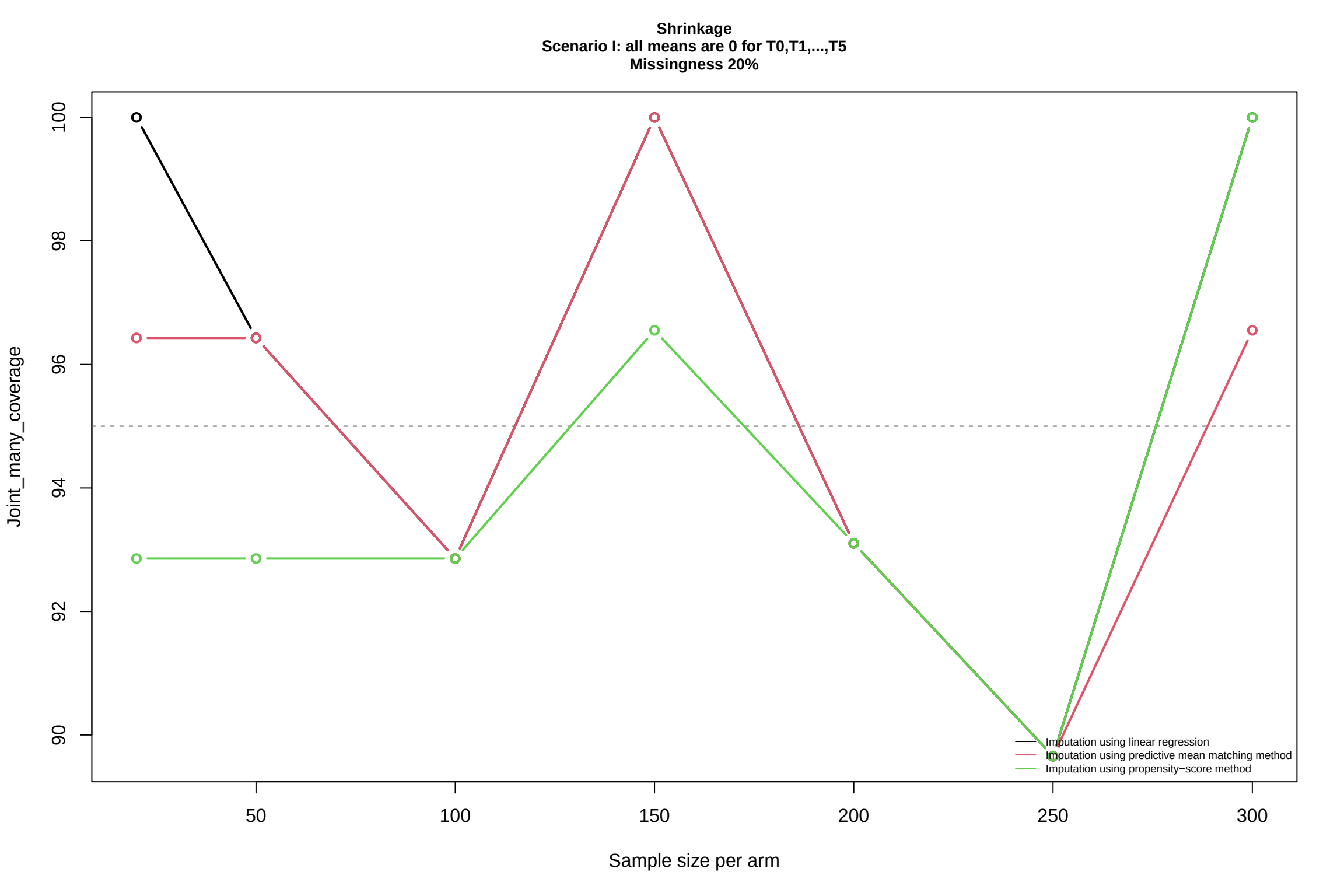

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 40%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

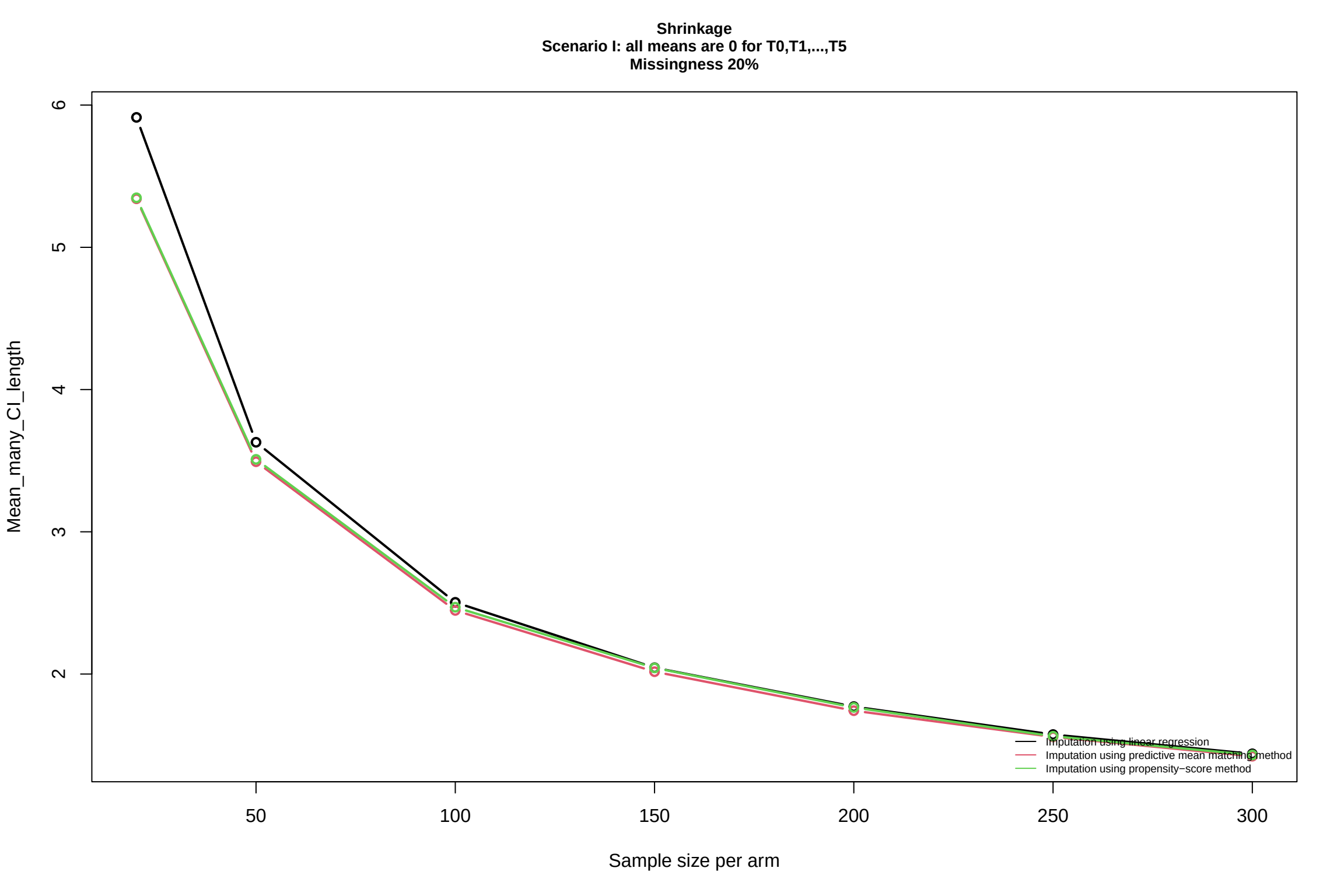

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 0%
Joint_many_coverage
100
98
96
94
92
90
50
100
150
200
250
300
Sample size per arm
Complete data

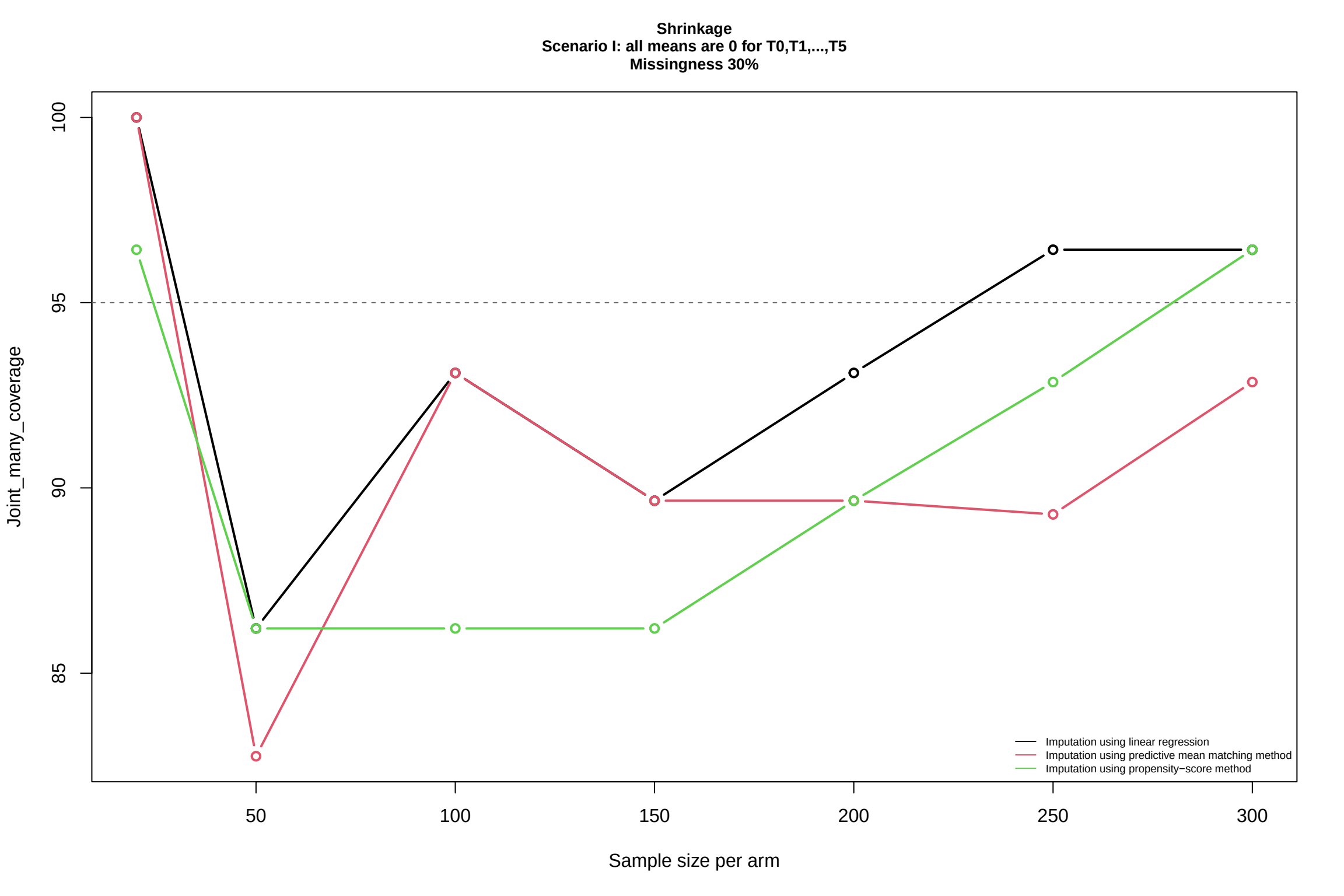

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

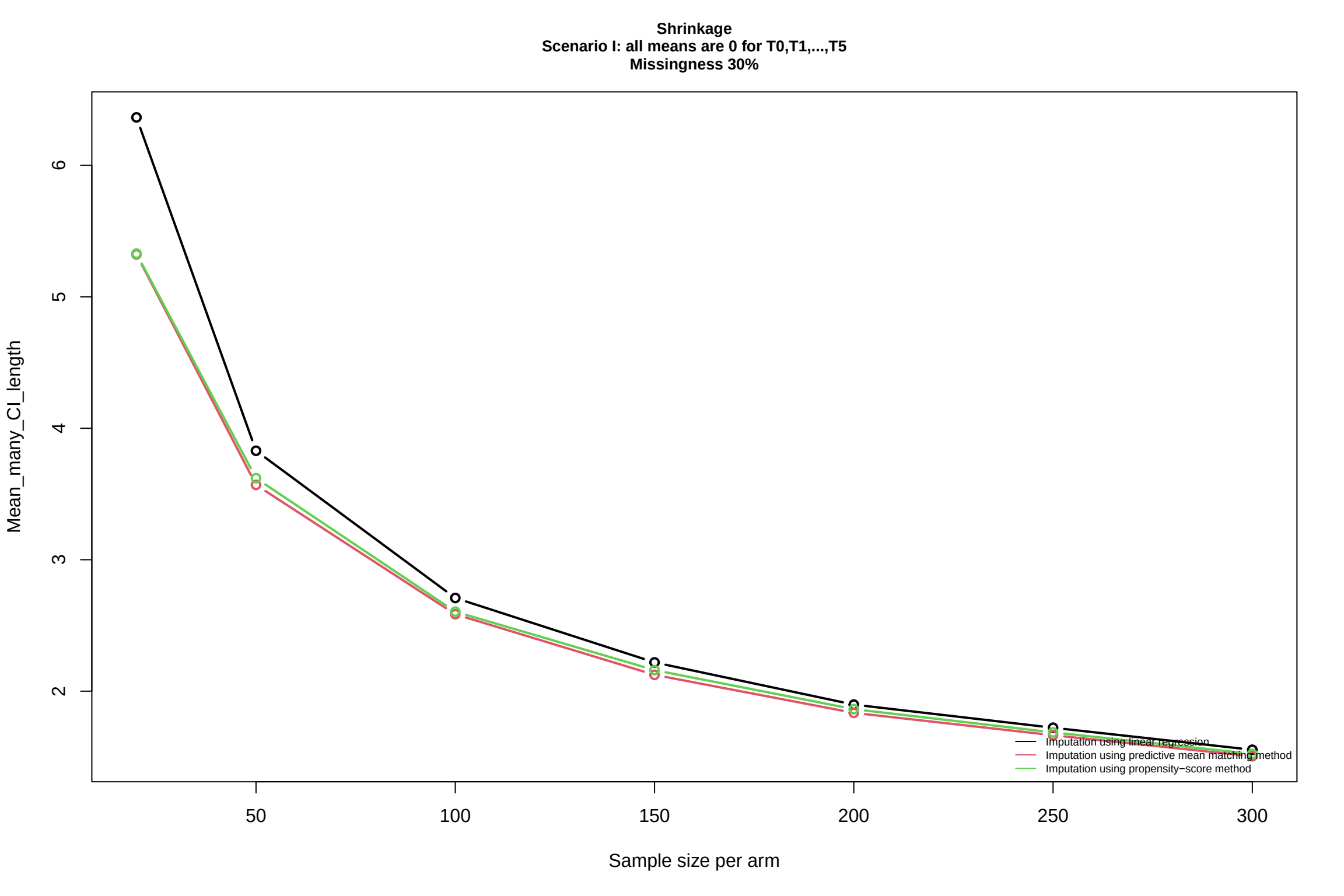

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 10%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

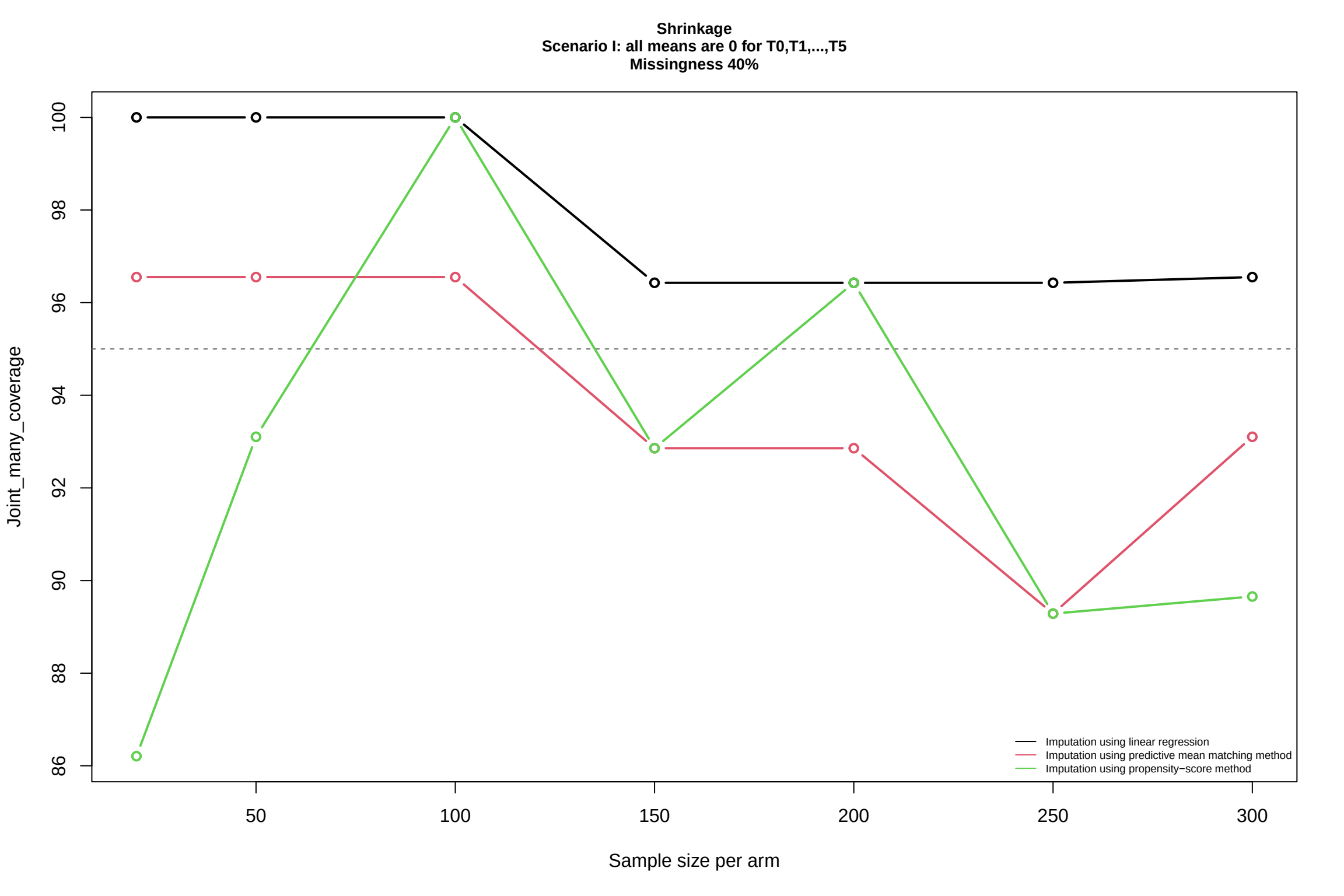

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 10%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

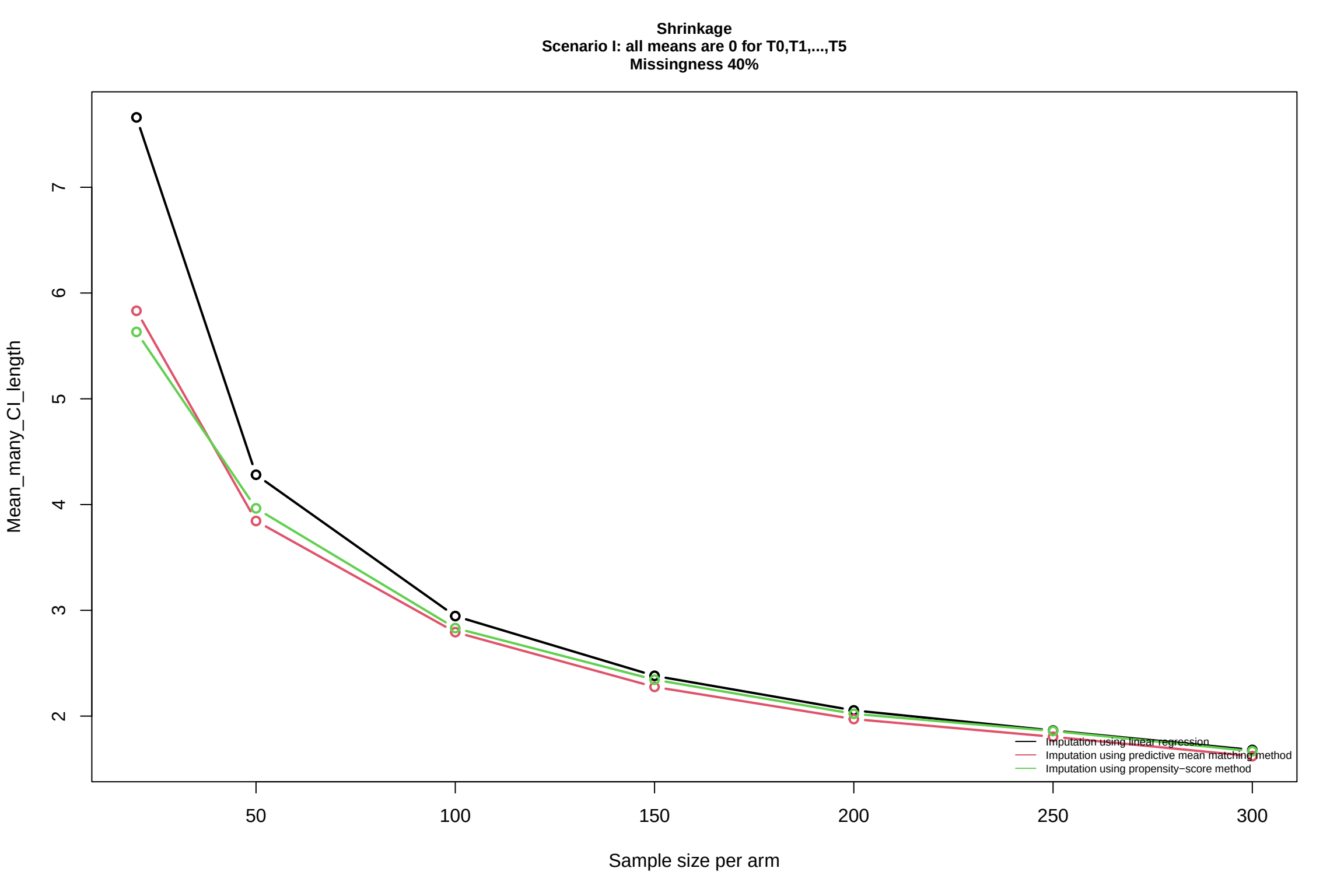

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 20%
Joint_many_coverage
100
98
96
94
92
90
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

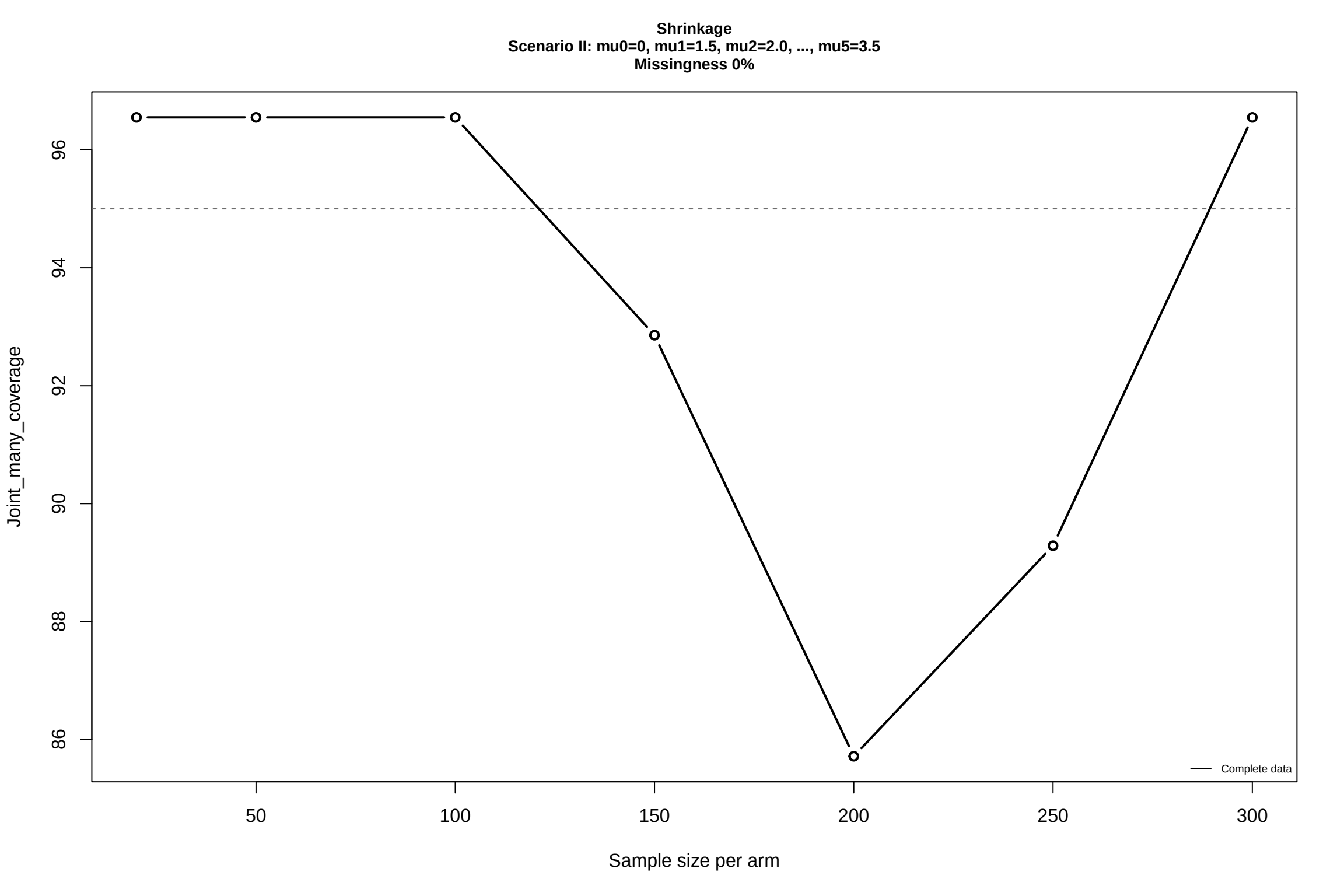

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 20%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

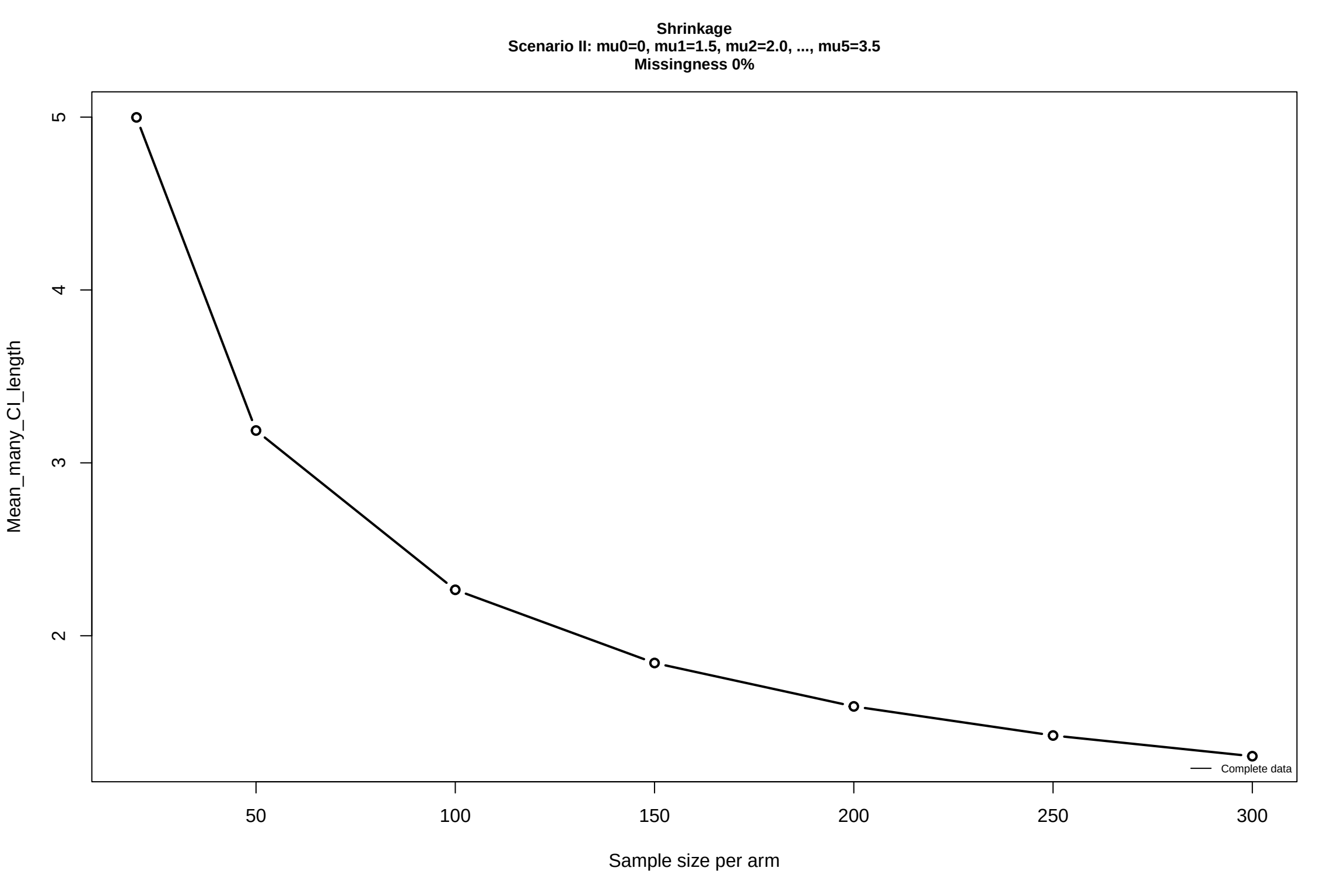

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 30%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

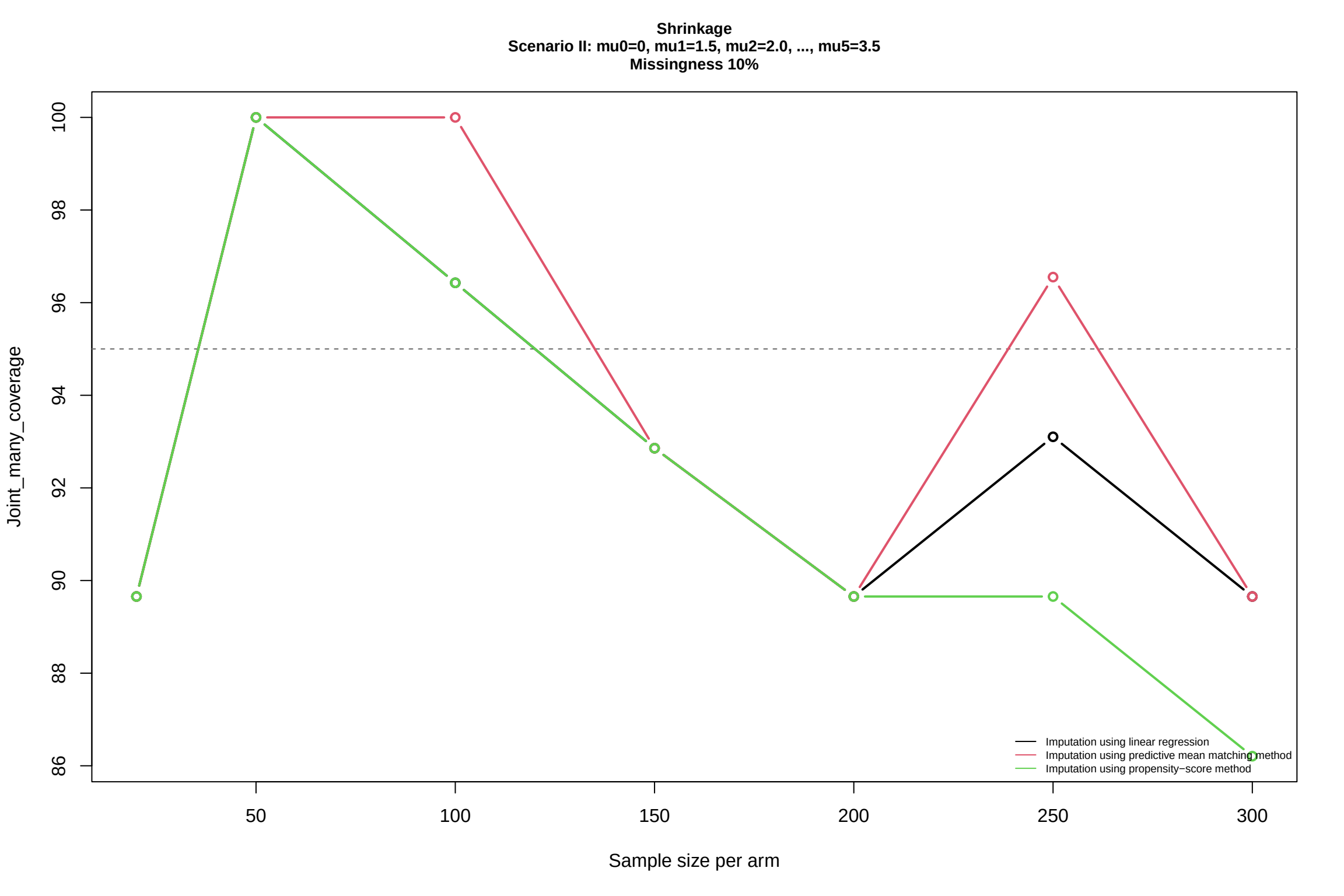

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 30%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

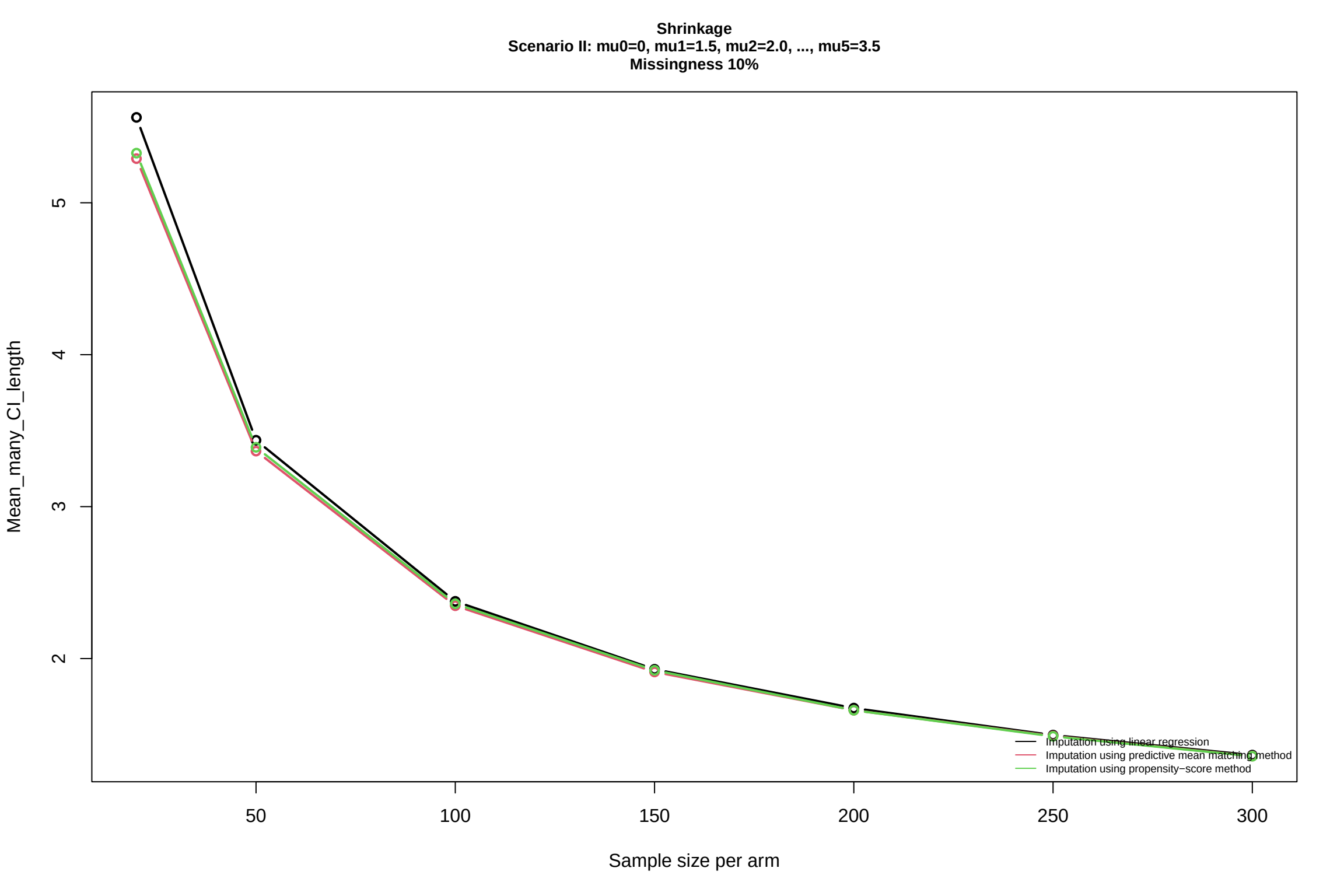

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 40%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

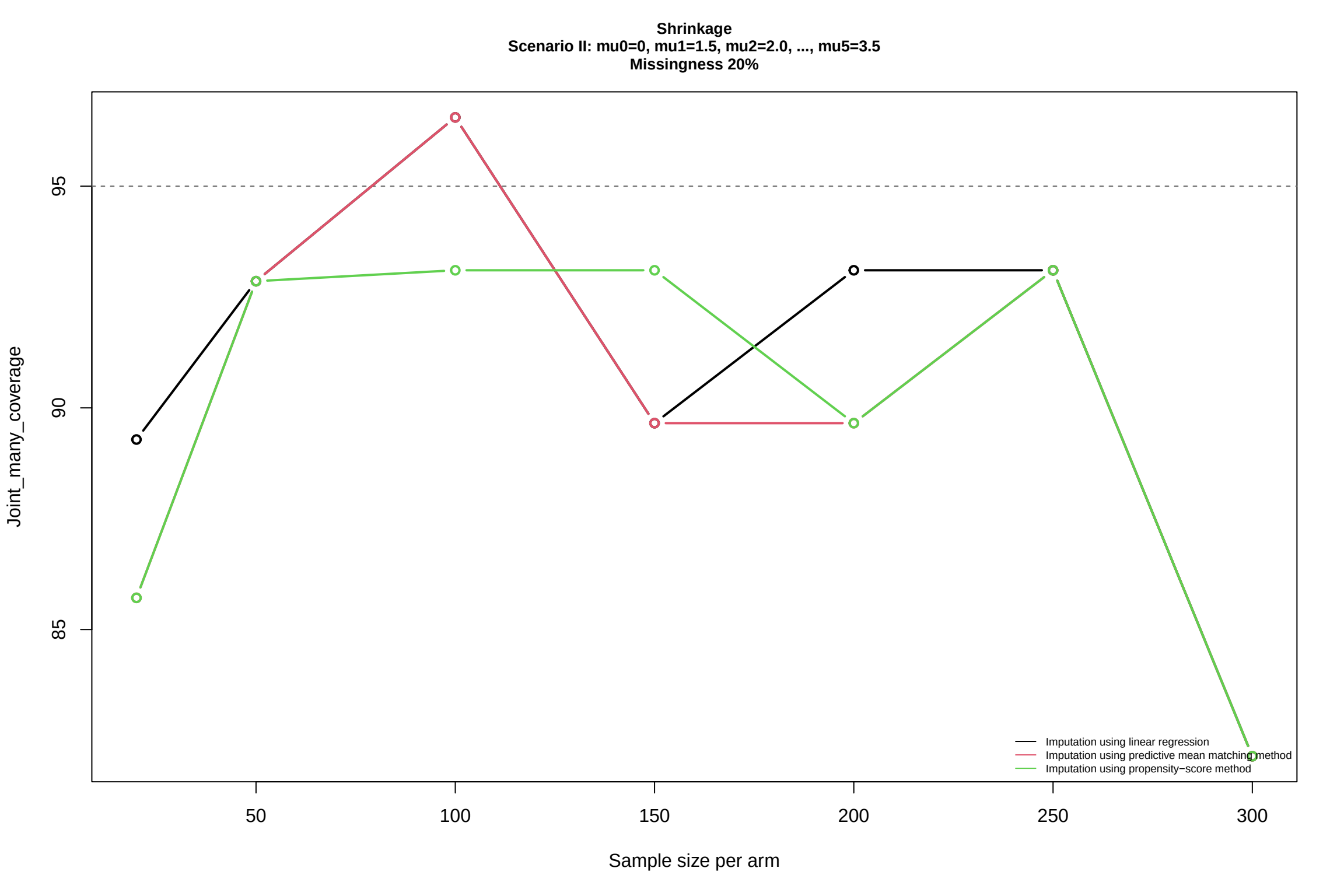

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 40%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

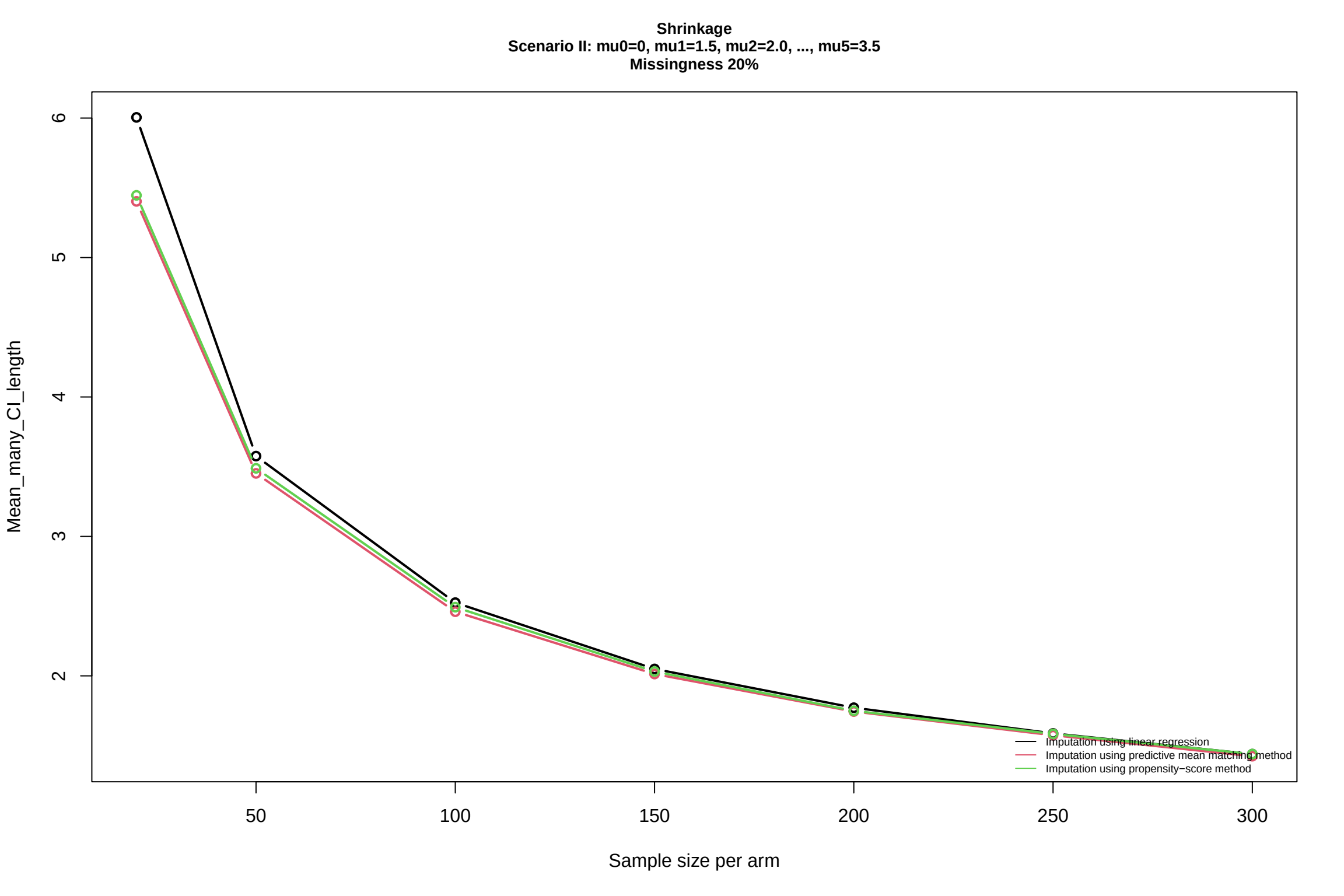

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 0%
Joint_many_coverage
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Complete data

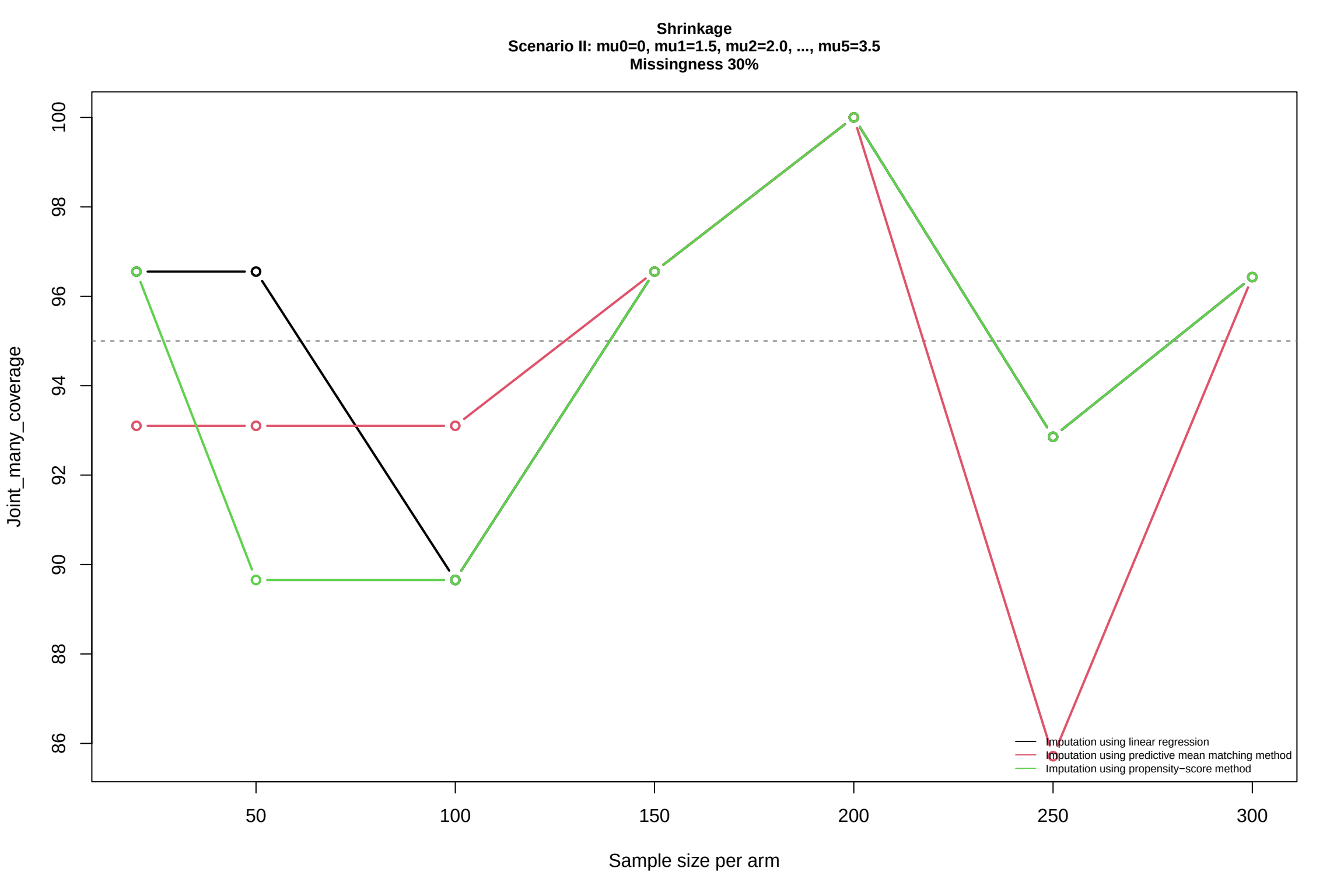

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

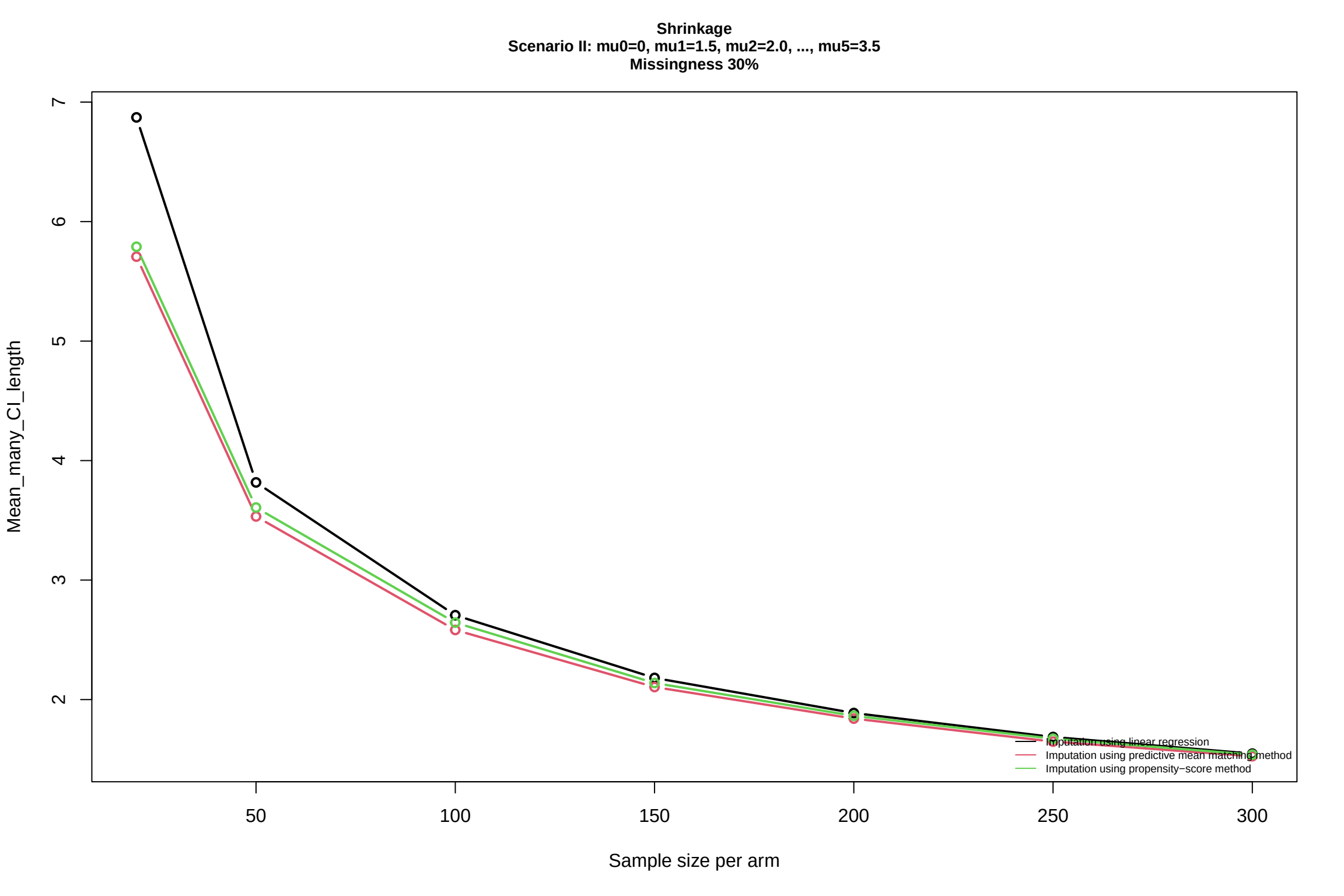

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 10%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

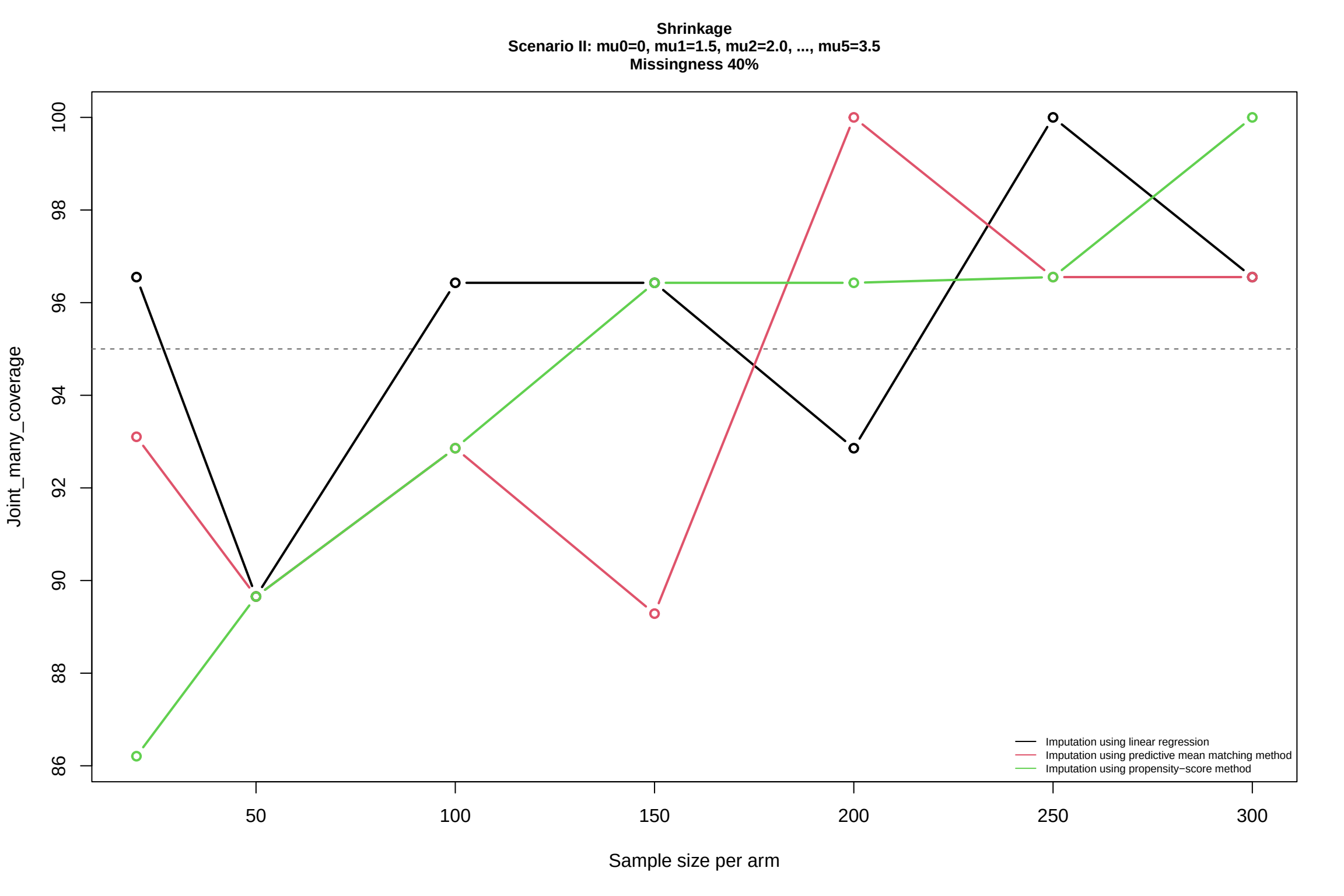

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 10%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

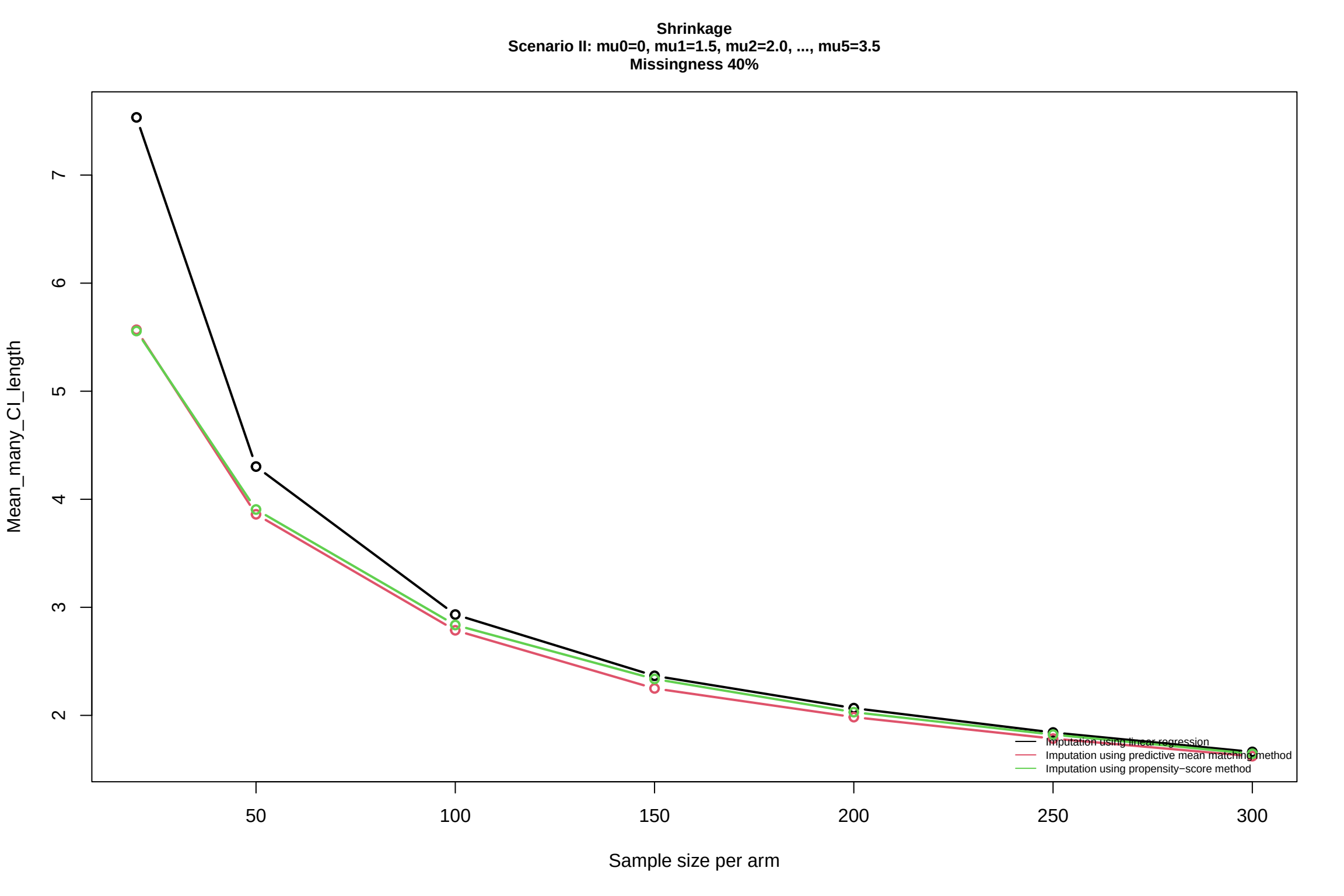

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 20%
Joint_many_coverage
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 20%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 30%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 30%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 40%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 40%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method